

Comparing Optimization Algorithms

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Outline



1. Introduction
2. Views on Performance and Time
3. Statistical Measures
4. Statistical Comparisons
5. Testing is Not Enough
6. Other Stuff
7. Summary
8. Advertisement





Introduction



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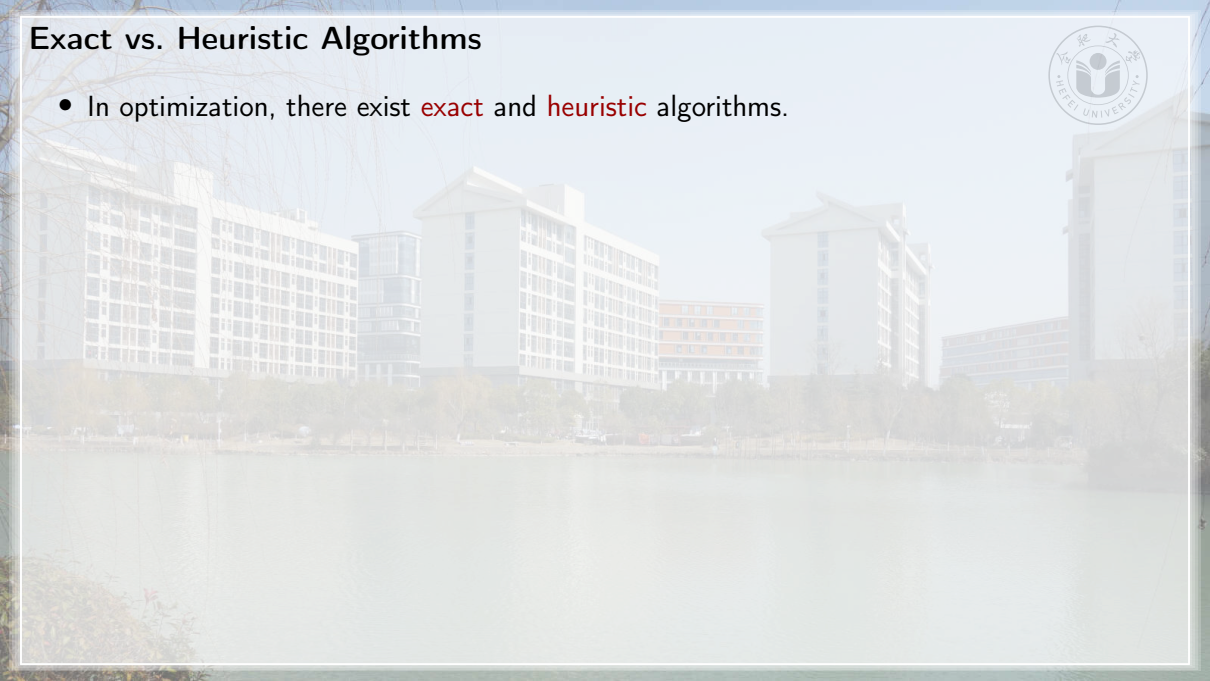
Introduction



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- For solving an optimization problem, we want to use the algorithm most suitable for it.
- What does this mean?
- And how do we find this algorithm?
- Hopefully this lesson will help answering these questions.
- As a complement to this lesson, I suggest the report *“Benchmarking in Optimization: Best Practice and Open Issues”*⁷ on arXiv.

Exact vs. Heuristic Algorithms

- In optimization, there exist **exact** and **heuristic** algorithms.



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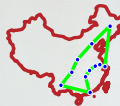


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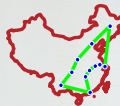


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- Let's look at the classical Traveling Salesperson Problem (TSP)^{3,54,73,75,124,131}.
 - Clearly, there is (at least) one shortest tour.
 - Theory proofs that the worst-case runtime needed to find this tour grows exponentially with the number n of cities we want to visit.^{1,23,26,65,72}

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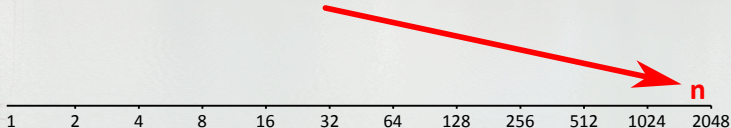


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- What does **exponential growth** mean?
- Let's say we have a number of cities n .

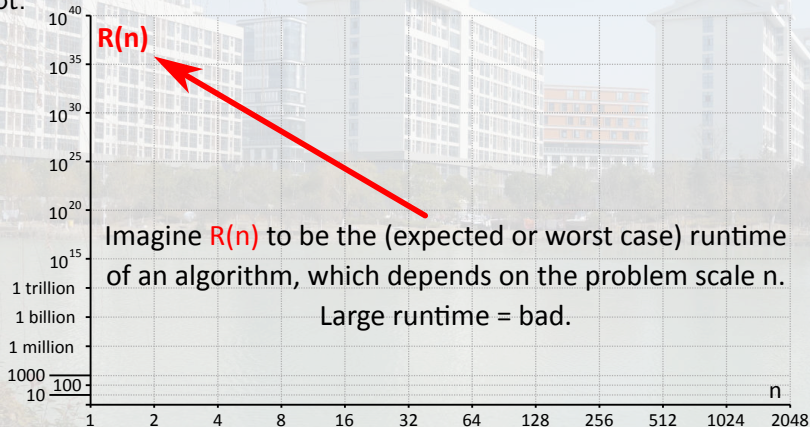
The problem scale n usually corresponding to the number of decision variables.



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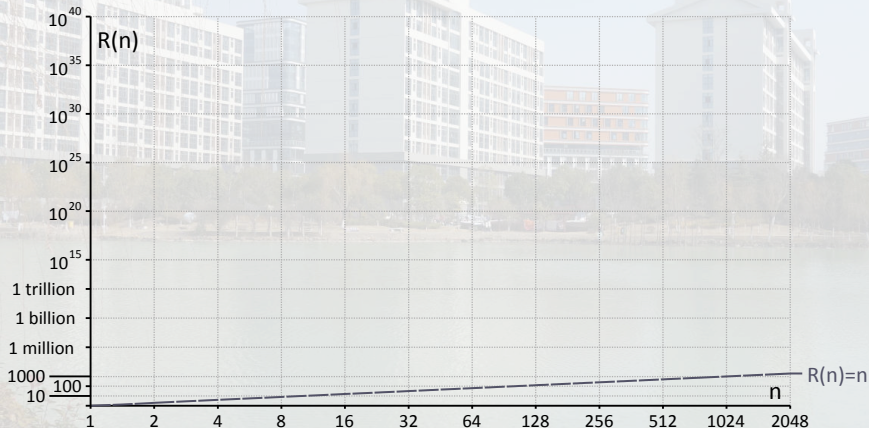
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- Let's say we have a number of cities n and a runtime as a function $R(n)$ in this log-log plot.



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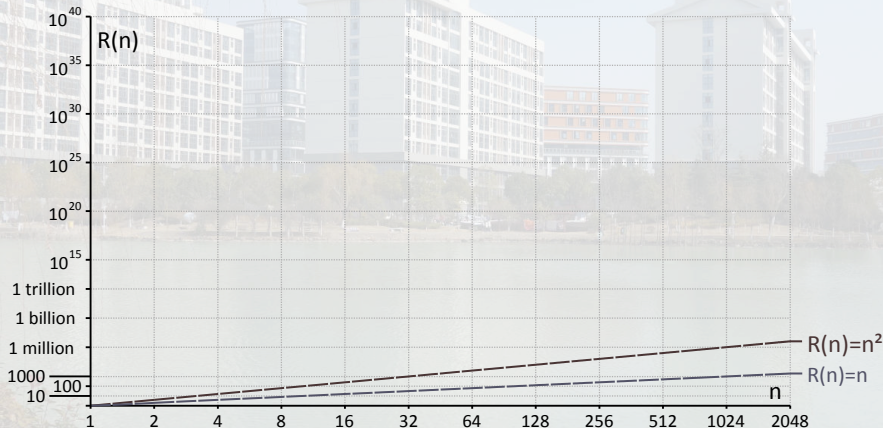
- In optimization, there exist **exact** and **heuristic** algorithms.
- Let's look at the classical Traveling Salesperson Problem (TSP)^{3,54,73,75,124,131}.
- A linear function means that the runtime $R(n)$ grows slowly with n .



Exact vs. Heuristic Algorithms



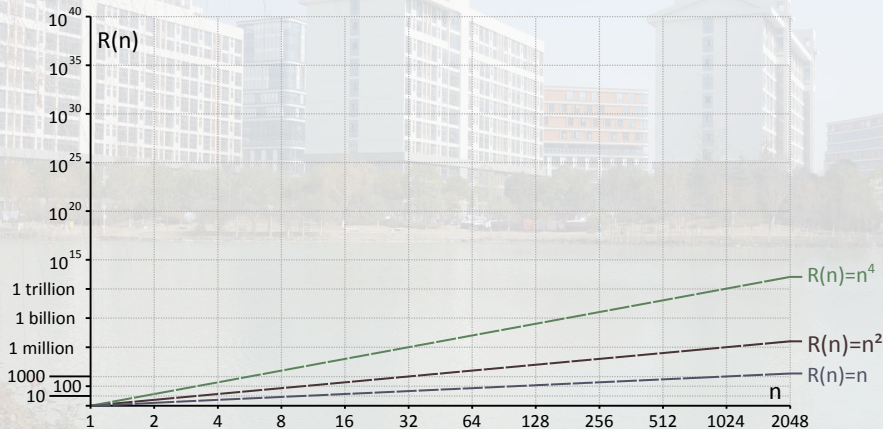
- In optimization, there exist **exact** and **heuristic** algorithms.
- Let's look at the classical Traveling Salesperson Problem (TSP)^{3,54,73,75,124,131}.
- A quadratic function (a straight line in log-log plots) is also OK.



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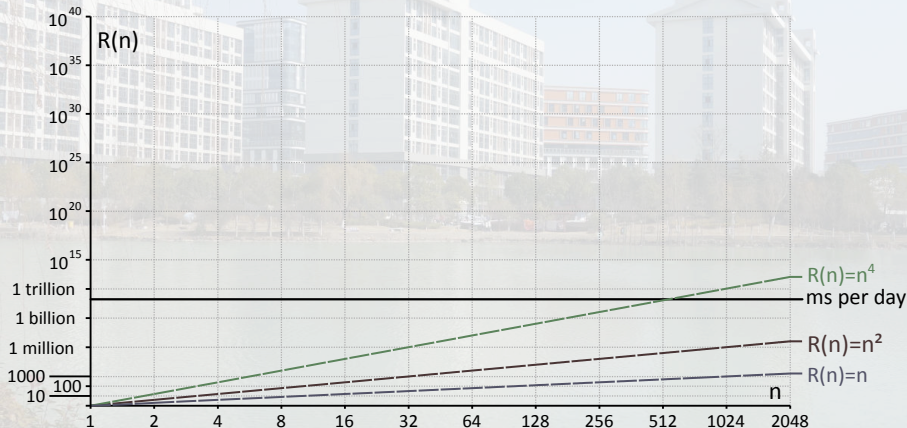
- In optimization, there exist **exact** and **heuristic** algorithms.
- Let's look at the classical Traveling Salesperson Problem (TSP)^{3,54,73,75,124,131}.
- A quartic function $R(n) = n^4$ gets quite large for growing n .



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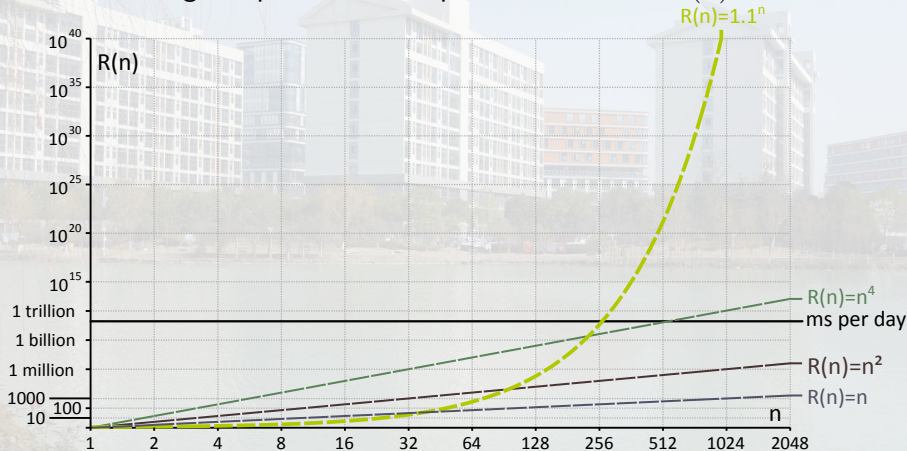
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- A quartic function exceeds the number of milliseconds per day at $n \approx 512$.



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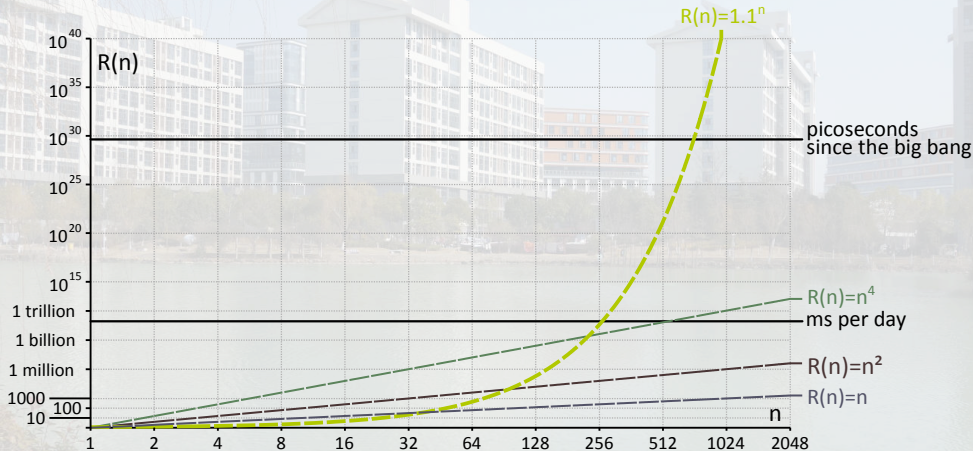
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- But this is nothing compared to the exponential function $R(n) = 1.1^n \dots$



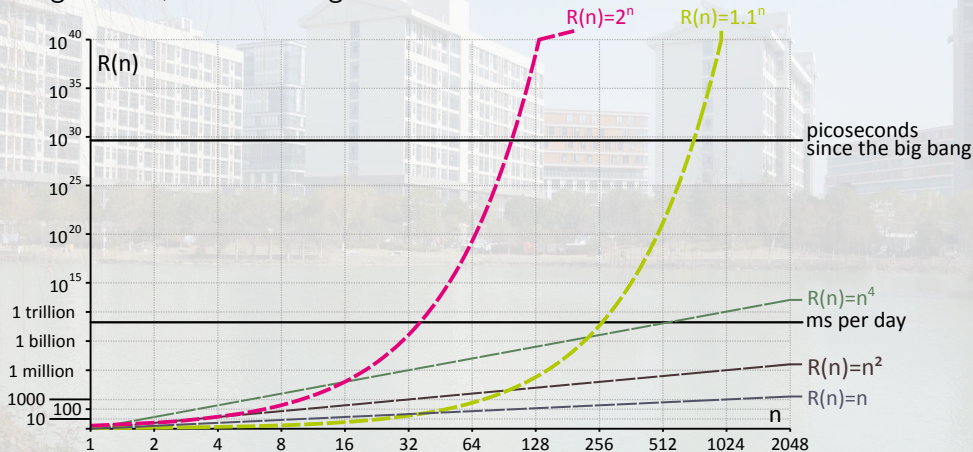
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- A runtime of 1.1^n becomes infeasible for $n > 512$.



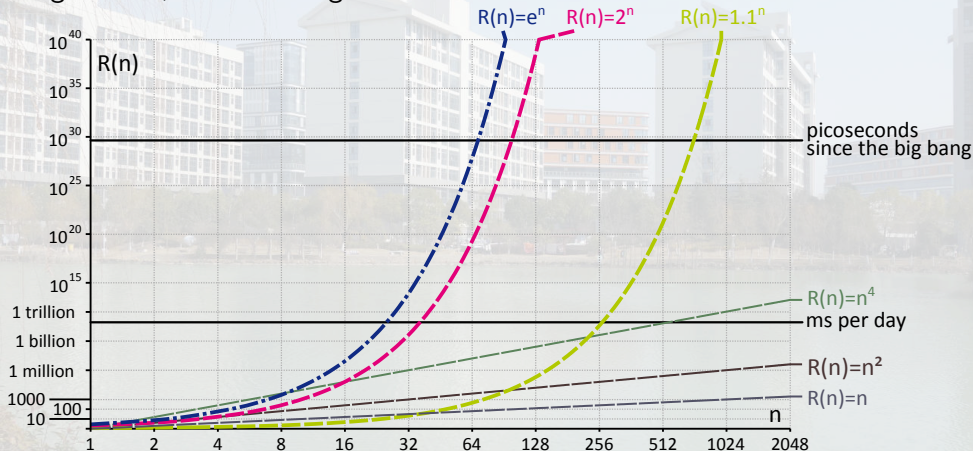
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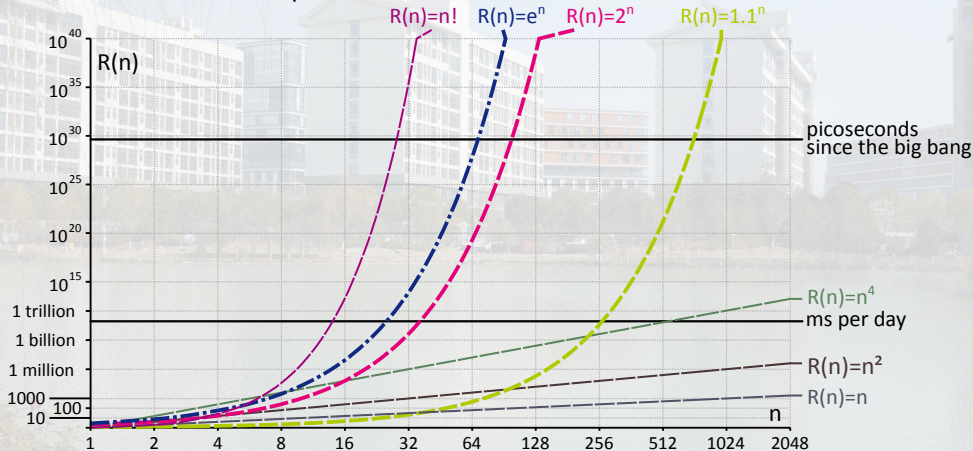
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- For larger bases, the runtime grows even faster.



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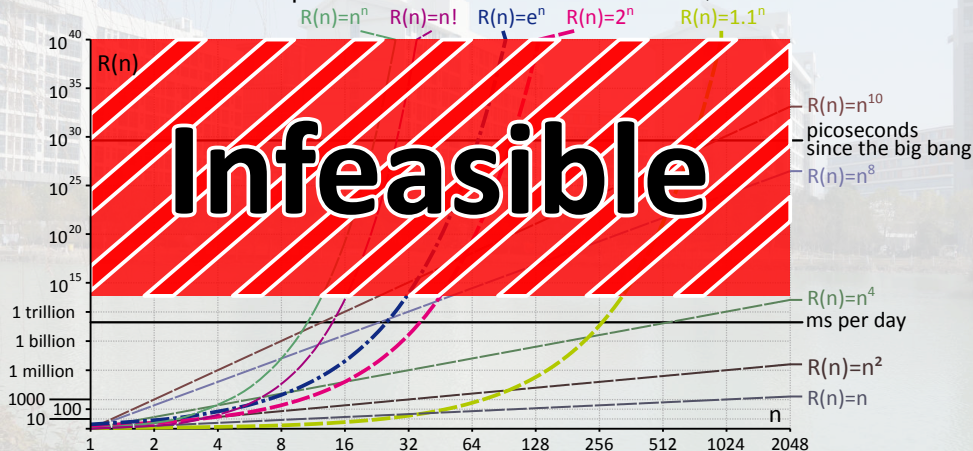
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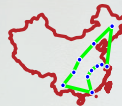


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very much / too (?) long

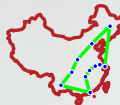
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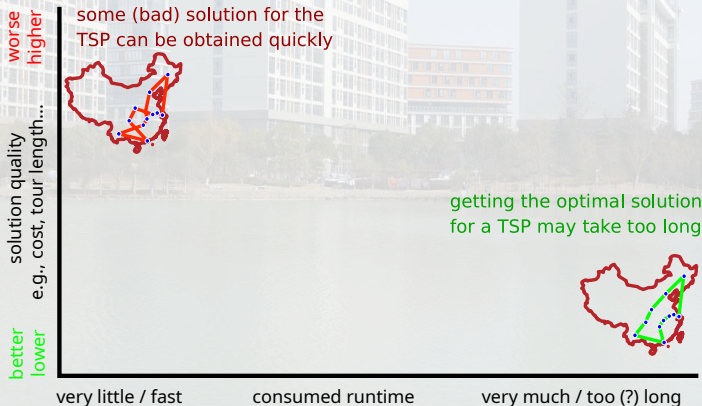
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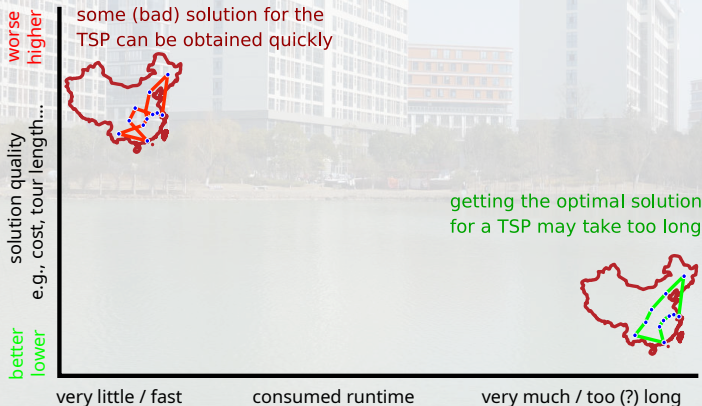
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 - But we can find just **some** tour very quickly.
 - Of course the quality of that tour will be lower: the tour will be longer than the best one.



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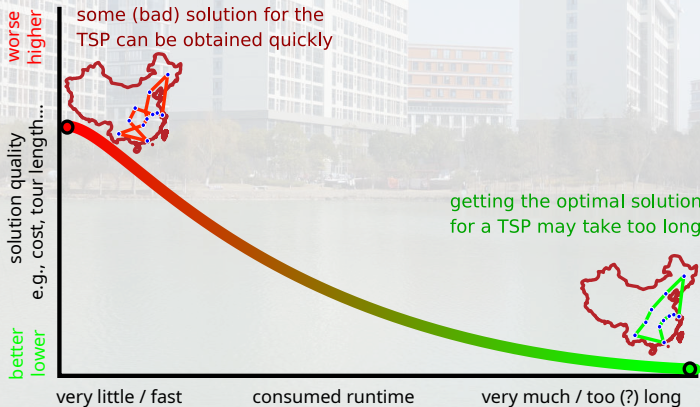
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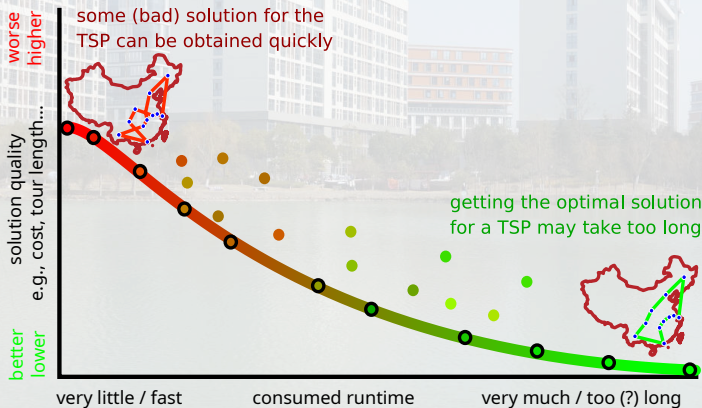
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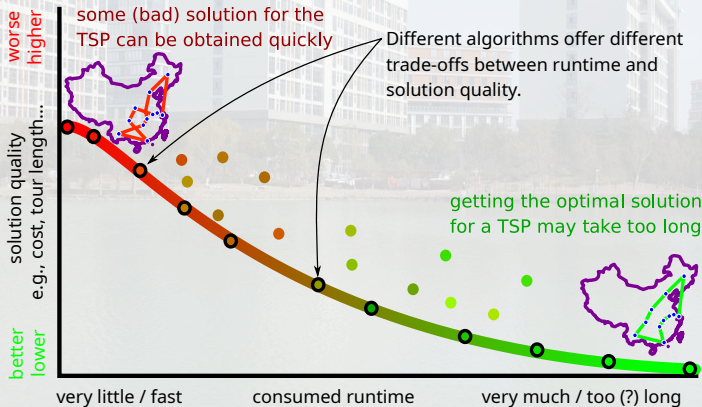
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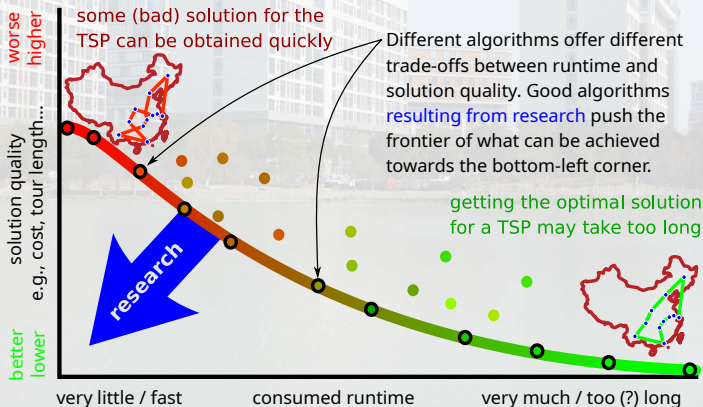
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Views on Performance and Time



Views on Performance



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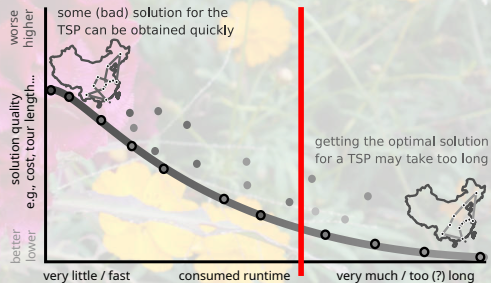


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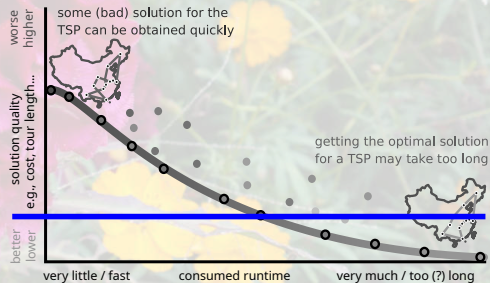
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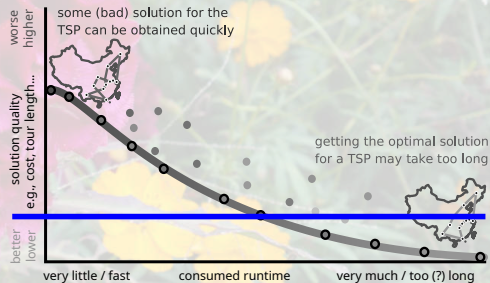
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What is Runtime?



- What actually is **runtime**?

Clock Time as Absolute Runtime

We can measure the (absolute) consumed runtime of the algorithm in ms.



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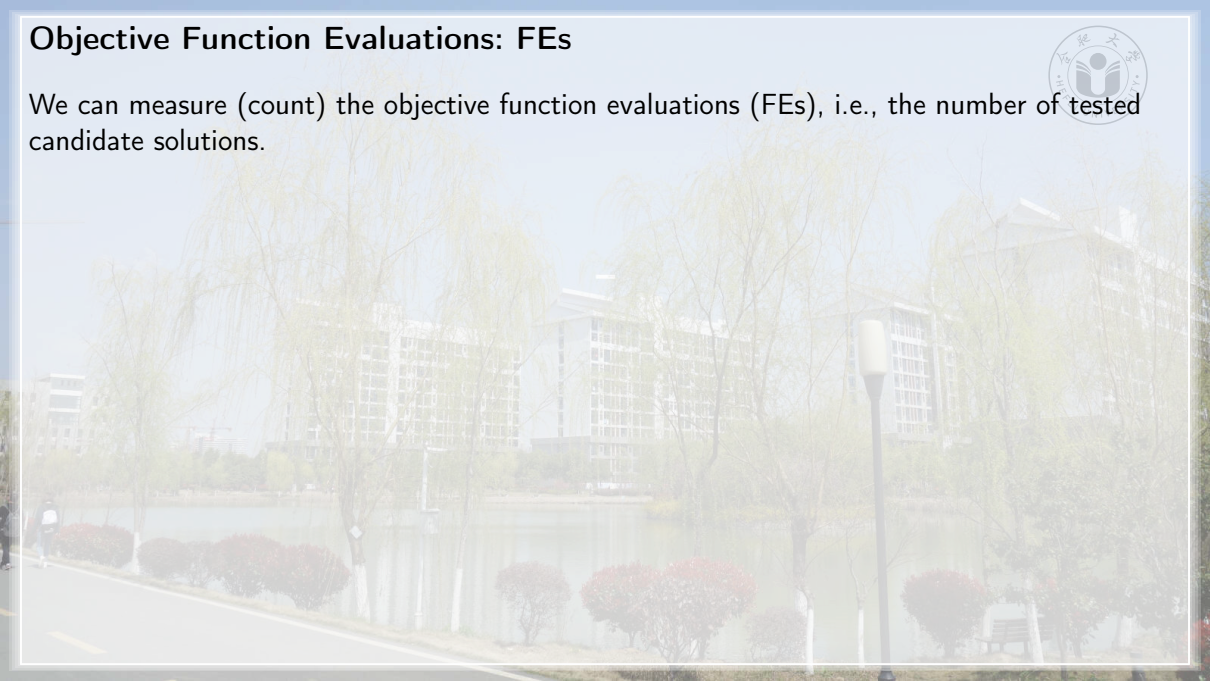
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- ...for **research** they may be less interesting, while for a **specific application** they do matter.

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We can measure (count) the objective function evaluations (FEs), i.e., the number of tested candidate solutions.

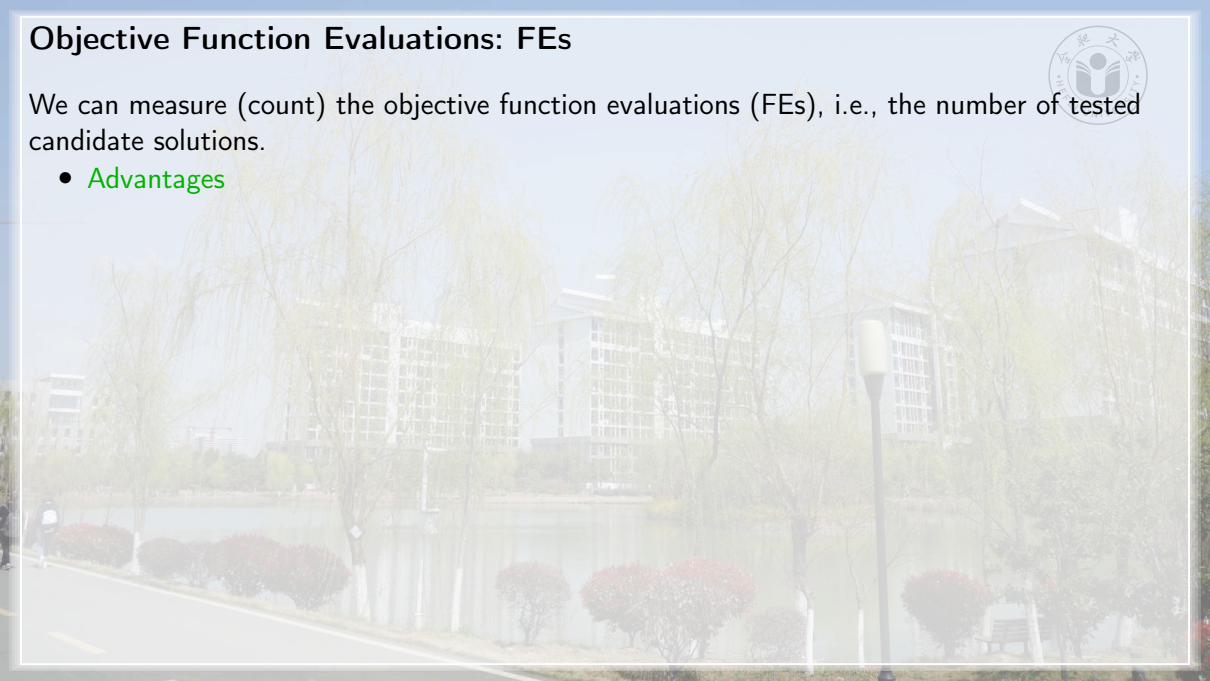


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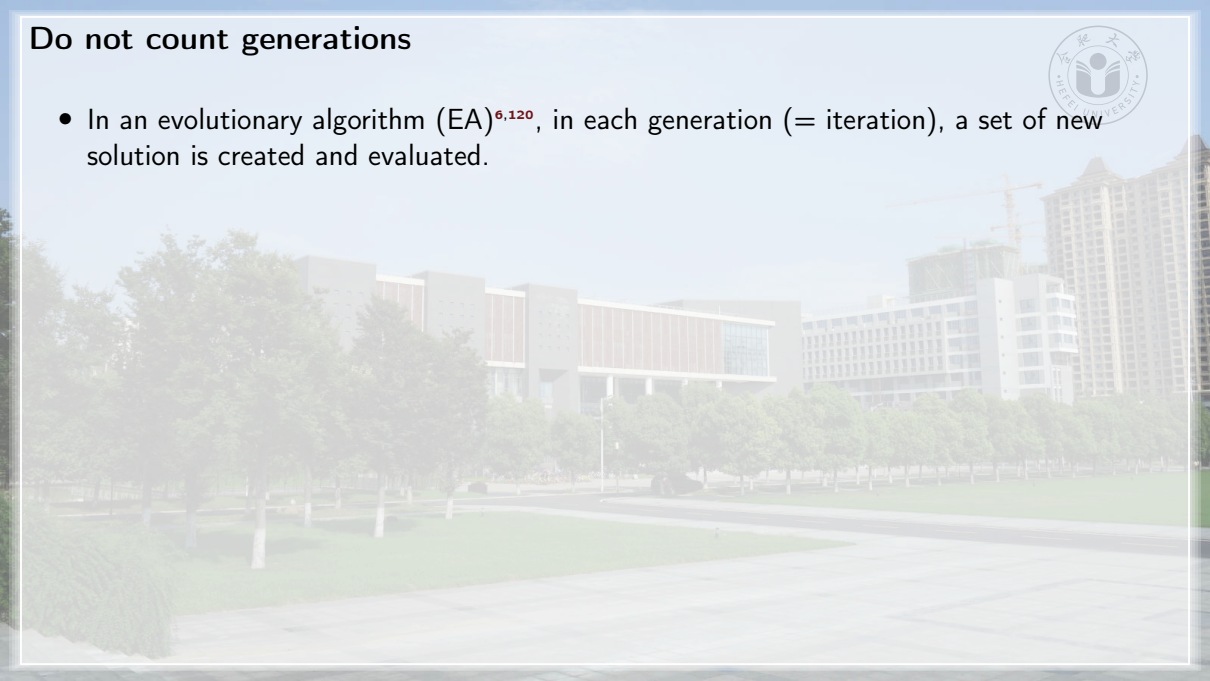
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 - When applying Ant Colony Optimization (ACO) instead, each FE takes $\mathcal{O}(n^2)$ ¹²⁴.
- Relevant for comparing algorithms, but not so much for the practical application or comparing implementations.

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- Do not use the number of *generations* in your EA as time measure! Instead count the FEs, because:
- The “number of generations” are not really comparable for different population sizes or with algorithms that do not use populations.

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- I suggest to prefer FEs over generations if you want to count algorithm steps.

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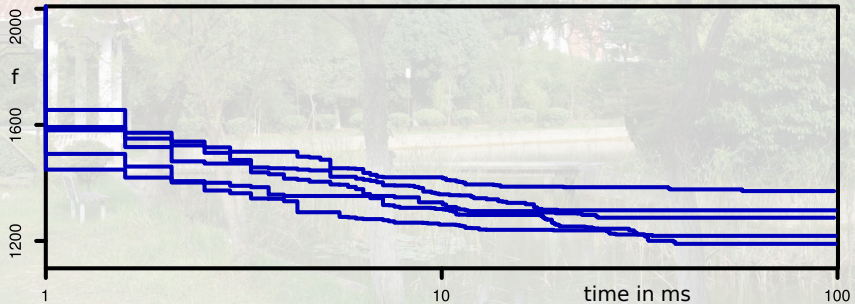


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Views on Performance



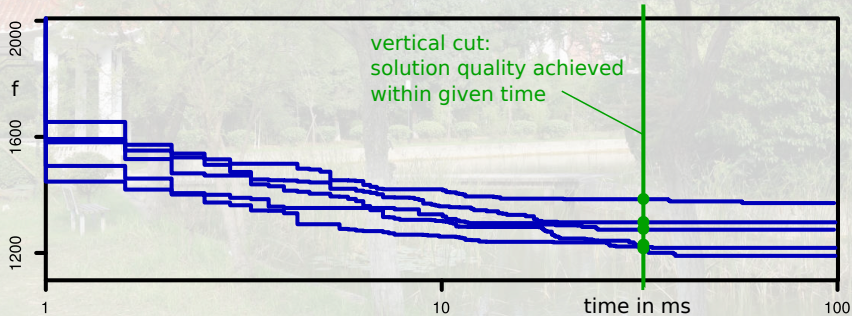
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Views on Performance



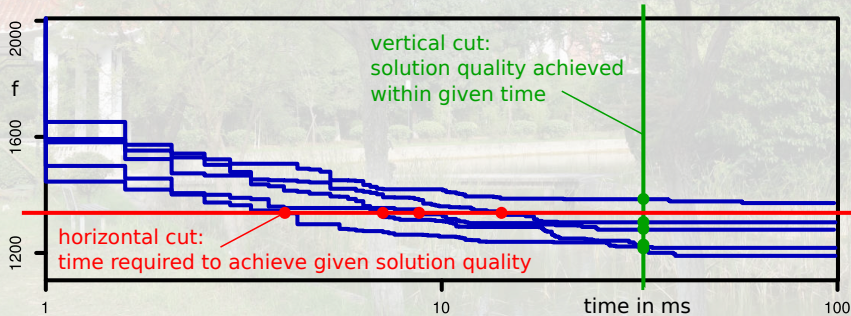
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Views on Performance



- Which one is the “better” view on performance?
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- This question is still debated in research. . .

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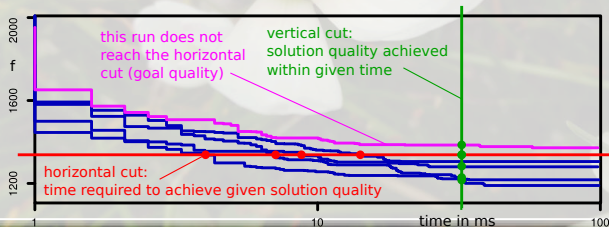


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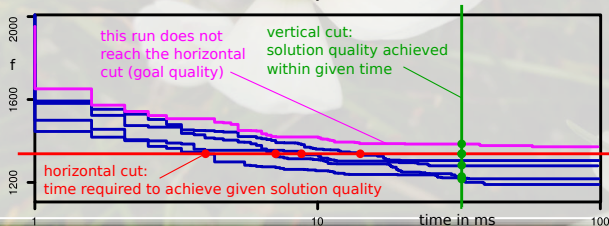
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- Then, alternative measures need to be computed, such as the ERT^{4,99} or PAR2 and PAR10^{12,66}.



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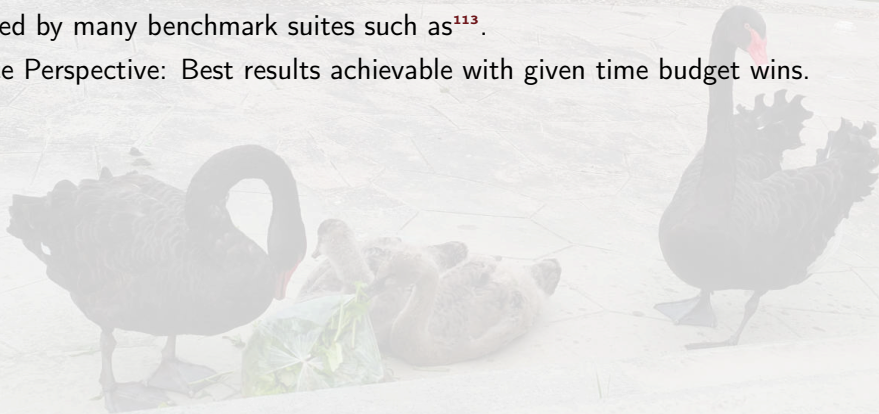
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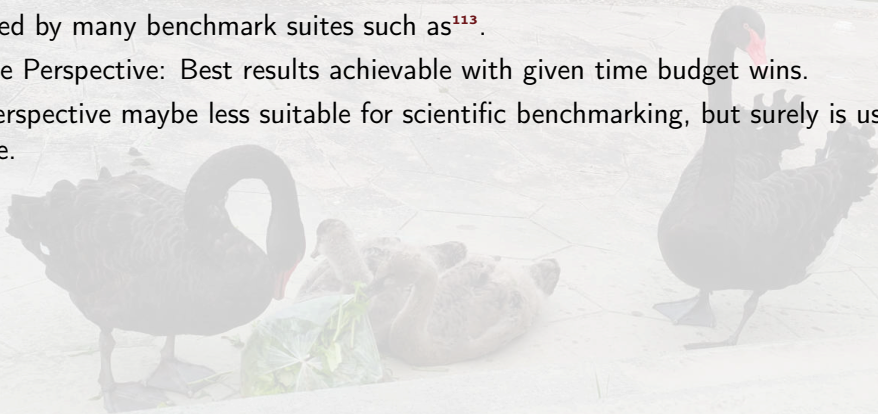
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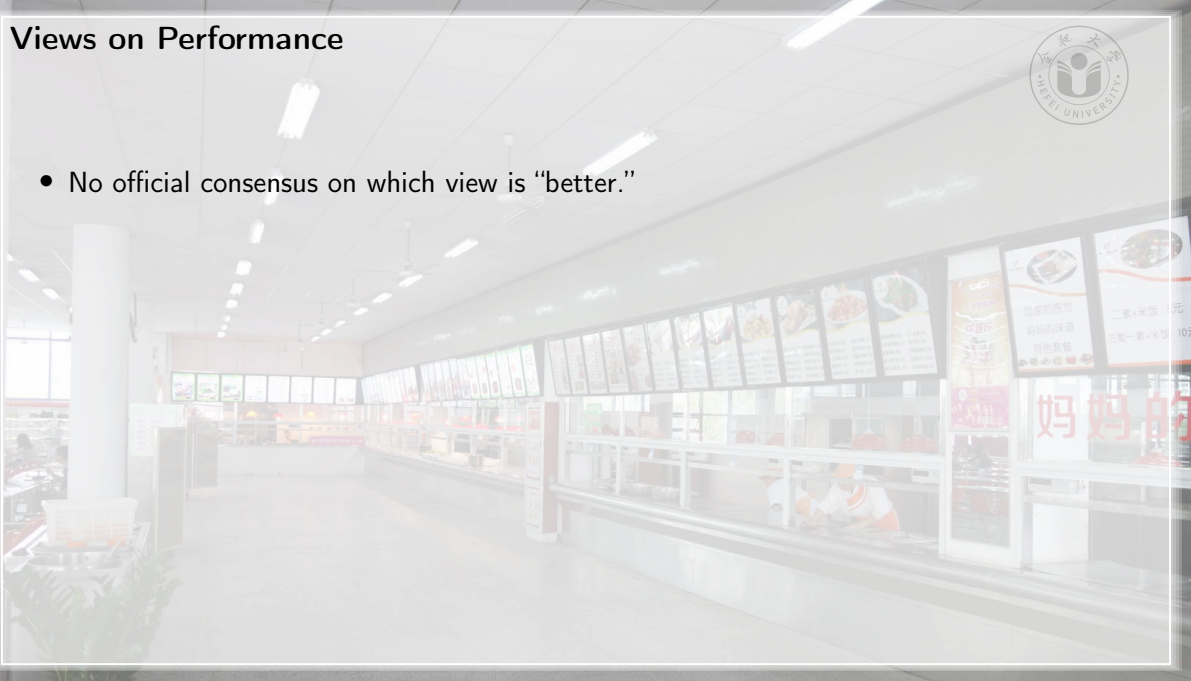


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- “How good is the tour for the TSP that we can find in 5 minutes with our algorithm?”
- Always well-defined, because vertical cuts can always be reached.

Views on Performance



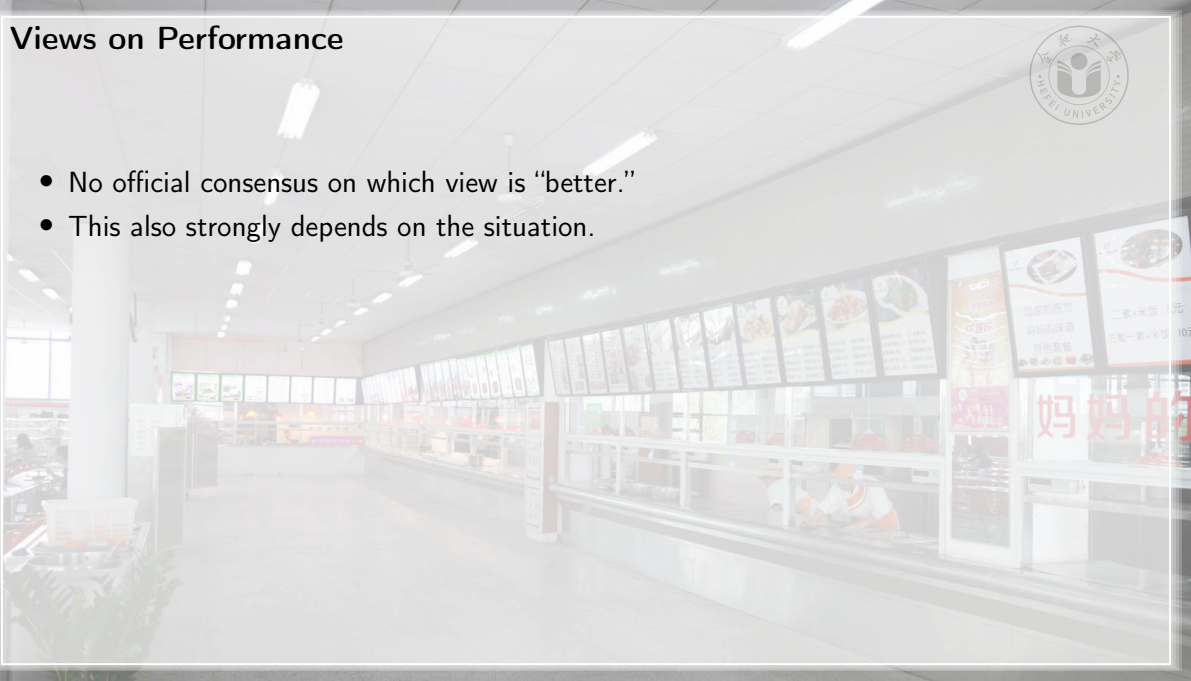
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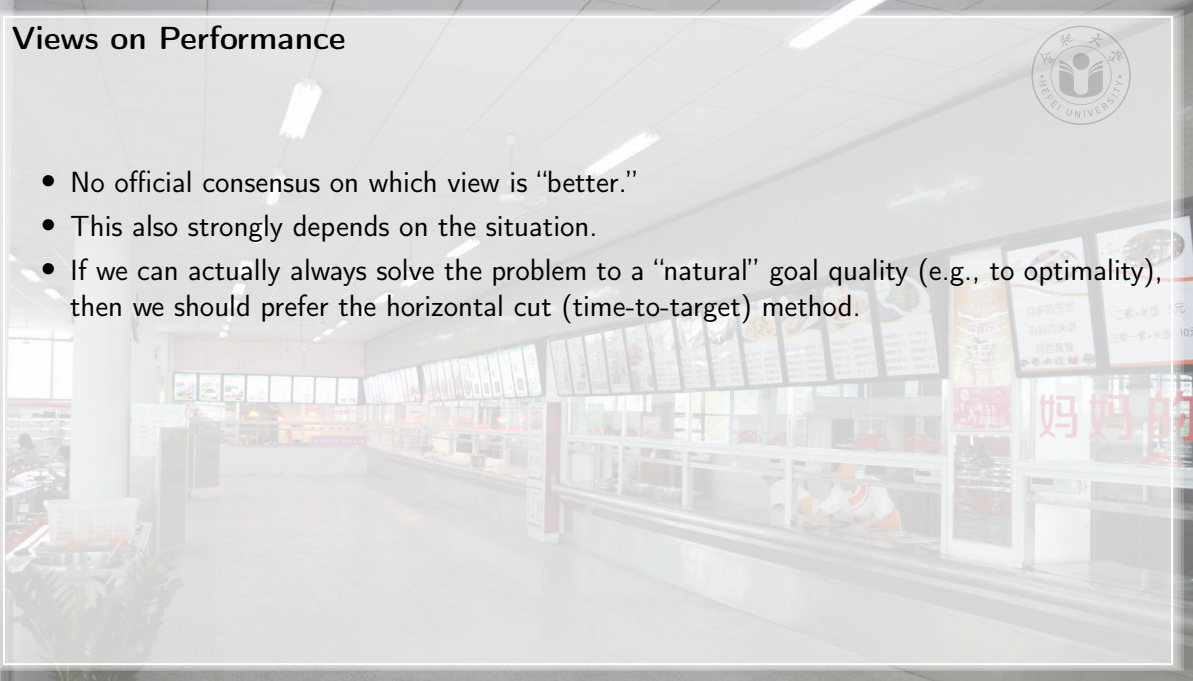
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- Maybe cast a net of several horizontal and vertical cuts, to get a better picture. . .

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Statistical Measures



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- Example: Dice Throw
- How likely is it to roll a 1, 2, 3, 4, 5, or 6?



Important Distinction



# throws	number	$f(1)$	$f(2)$	$f(3)$	$f(4)$	$f(5)$	$f(6)$
1	5	0.0000	0.0000	0.0000	0.0000	1.0000	0.0000



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2	4	0.0000	0.0000	0.0000	0.5000	0.5000	0.0000



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1	5	0.0000	0.0000	0.0000	0.0000	1.0000	0.0000
2	4	0.0000	0.0000	0.0000	0.5000	0.5000	0.0000
3	1	0.3333	0.0000	0.0000	0.3333	0.3333	0.0000



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3	1	0.3333	0.0000	0.0000	0.3333	0.3333	0.0000
4	4	0.2500	0.0000	0.0000	0.5000	0.2500	0.0000
5	3	0.2000	0.0000	0.2000	0.4000	0.2000	0.0000



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3	1	0.3333	0.0000	0.0000	0.3333	0.3333	0.0000
4	4	0.2500	0.0000	0.0000	0.5000	0.2500	0.0000
5	3	0.2000	0.0000	0.2000	0.4000	0.2000	0.0000
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6	3	0.1667	0.0000	0.3333	0.3333	0.1667	0.0000
7	2	0.1429	0.1429	0.2857	0.2857	0.1429	0.0000
8	1	0.2500	0.1250	0.2500	0.2500	0.1250	0.0000
9	4	0.2222	0.1111	0.2222	0.3333	0.1111	0.0000
10	2	0.2000	0.2000	0.2000	0.3000	0.1000	0.0000



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9	4	0.2222	0.1111	0.2222	0.3333	0.1111	0.0000
10	2	0.2000	0.2000	0.2000	0.3000	0.1000	0.0000
11	6	0.1818	0.1818	0.1818	0.2727	0.0909	0.0909
12	3	0.1667	0.1667	0.2500	0.2500	0.0833	0.0833



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11	6	0.1818	0.1818	0.1818	0.2727	0.0909	0.0909
12	3	0.1667	0.1667	0.2500	0.2500	0.0833	0.0833
100	...	0.1900	0.2100	0.1500	0.1600	0.1200	0.1700
1'000	...	0.1700	0.1670	0.1620	0.1670	0.1570	0.1770
10'000	...	0.1682	0.1699	0.1680	0.1661	0.1655	0.1623
100'000	...	0.1671	0.1649	0.1664	0.1676	0.1668	0.1672
1'000'000	...	0.1673	0.1663	0.1662	0.1673	0.1666	0.1664



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# throws	number	f(1)	f(2)	f(3)	f(4)	f(5)	f(6)
1	5	0.0000	0.0000	0.0000	0.0000	1.0000	0.0000
2	4	0.0000	0.0000	0.0000	0.5000	0.5000	0.0000
3	1	0.3333	0.0000	0.0000	0.3333	0.3333	0.0000
4	4	0.2500	0.0000	0.0000	0.5000	0.2500	0.0000
5	3	0.2000	0.0000	0.2000	0.4000	0.2000	0.0000
6	3	0.1667	0.0000	0.3333	0.3333	0.1667	0.0000
7	2	0.1429	0.1429	0.2857	0.2857	0.1429	0.0000
8	1	0.2500	0.1250	0.2500	0.2500	0.1250	0.0000
9	4	0.2222	0.1111	0.2222	0.3333	0.1111	0.0000
10	2	0.2000	0.2000	0.2000	0.3000	0.1000	0.0000
11	6	0.1818	0.1818	0.1818	0.2727	0.0909	0.0909
12	3	0.1667	0.1667	0.2500	0.2500	0.0833	0.0833
100	...	0.1900	0.2100	0.1500	0.1600	0.1200	0.1700
1'000	...	0.1700	0.1670	0.1620	0.1670	0.1570	0.1770
10'000	...	0.1682	0.1699	0.1680	0.1661	0.1655	0.1623
100'000	...	0.1671	0.1649	0.1664	0.1676	0.1668	0.1672
1'000'000	...	0.1673	0.1663	0.1662	0.1673	0.1666	0.1664
10'000'000	...	0.1667	0.1667	0.1666	0.1668	0.1667	0.1665
100'000'000	...	0.1667	0.1666	0.1666	0.1667	0.1667	0.1667
1'000'000'000	...	0.1667	0.1667	0.1667	0.1667	0.1667	0.1667



Important Distinction



- Crucial Difference: distribution and sample
- A sample is what we *measure*.
- A distribution is the asymptotic result of the ideal process.
- Statistical parameters of the distribution can be estimated from a sample.
- Example: Dice Throw
- How likely is it to roll a 1, 2, 3, 4, 5, or 6?
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- The parameters of a random process cannot be measured directly, but only be **estimated** from multiple measures.



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- We usually want to reduce this set of numbers to a single value which can give us an impression of what the “average outcome” (or result quality is).
- Three of the most common options for doing so, for estimating the “center” of a distribution, are the **arithmetic mean**, the **median**, and the **geometric mean**.

Arithmetic Mean



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$$\text{mean}(A) = \frac{1}{n} \sum_{i=0}^{n-1} a_i \quad (1)$$

Sample Median



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$$\text{median}(A) = \begin{cases} a_{\frac{n-1}{2}} & \text{if } n \text{ is odd} \\ \frac{1}{2} (a_{\frac{n}{2}-1} + a_{\frac{n}{2}}) & \text{otherwise} \end{cases} \quad \text{if } a_{i-1} \leq a_i \forall i \in 1 \dots (n-1) \quad (2)$$

Outliers

- Sometimes the data contains outliers^{53,82}.



Outliers

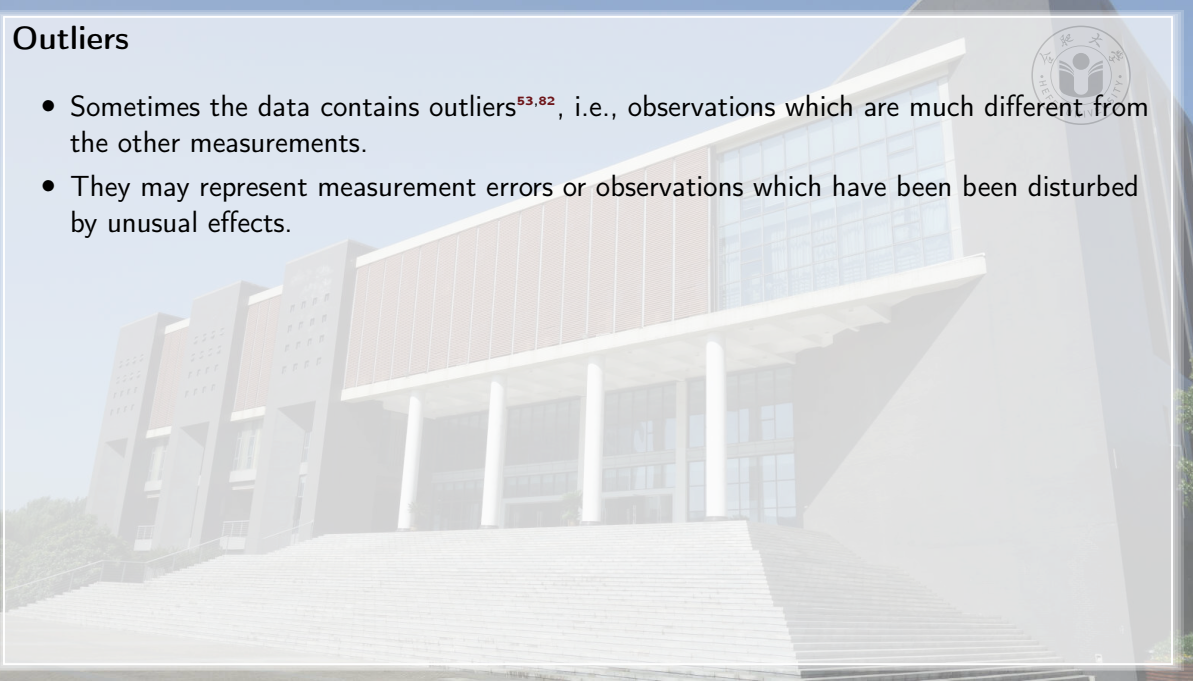
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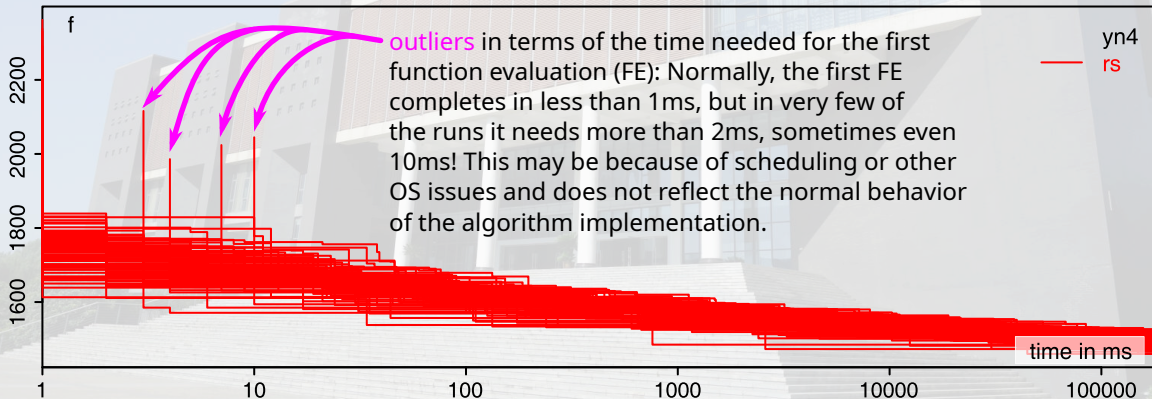


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- In my experiments here, there are sometimes outliers in the time that it takes to create and evaluate the first candidate solution.
- But outliers are actually important. So I say this right now. I will also say it again later. But I am afraid that you may tune out during the following example. So remember: Outliers are important. Anyway...

Example for Data Samples w/o Outlier



- Two sets of data samples A and B with $n_a = n_b = 19$ values.

$A = (1, 3, 4, 4, 4, 5, 6, 6, 6, 6, 7, 7, 9, 9, 9, 10, 11, 12, 14)$

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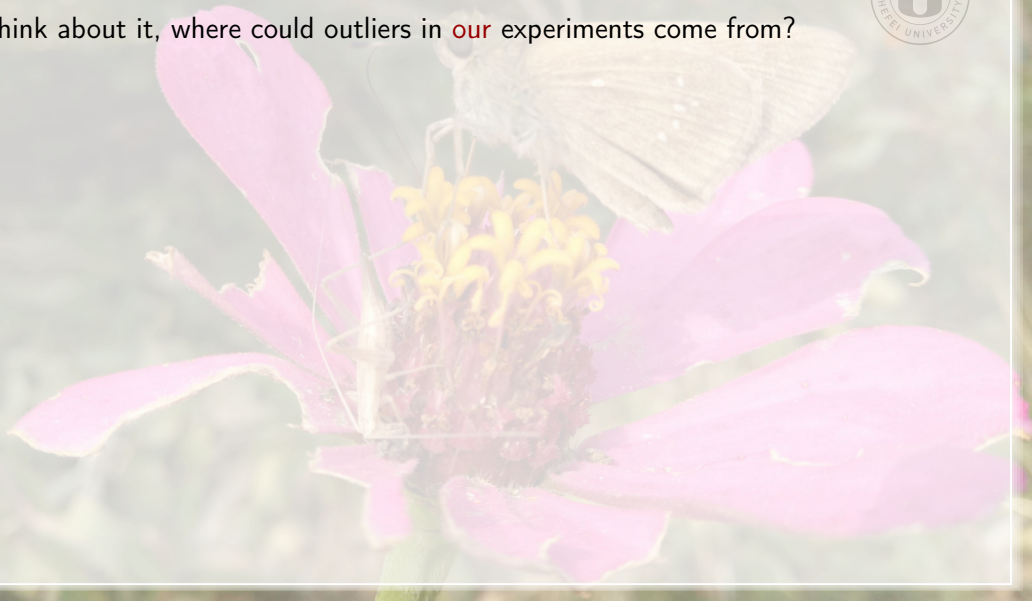
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- The median is not affected by the outliers.
- $\text{mean}(B) = 553$ is a value completely different from anything that actually occurs in B . . .
... it gives us a completely wrong impression.

Outliers can be important!

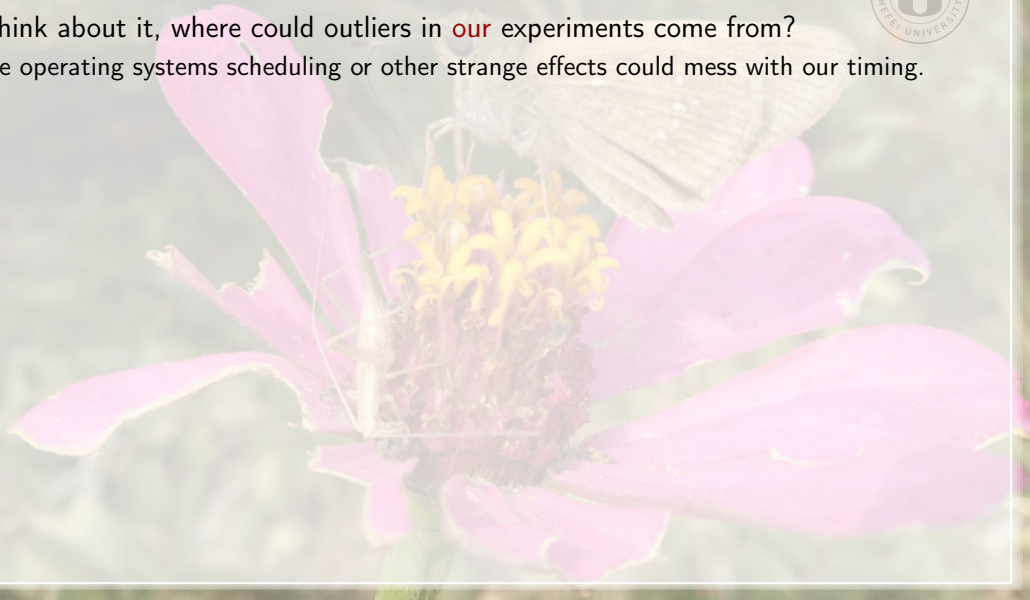
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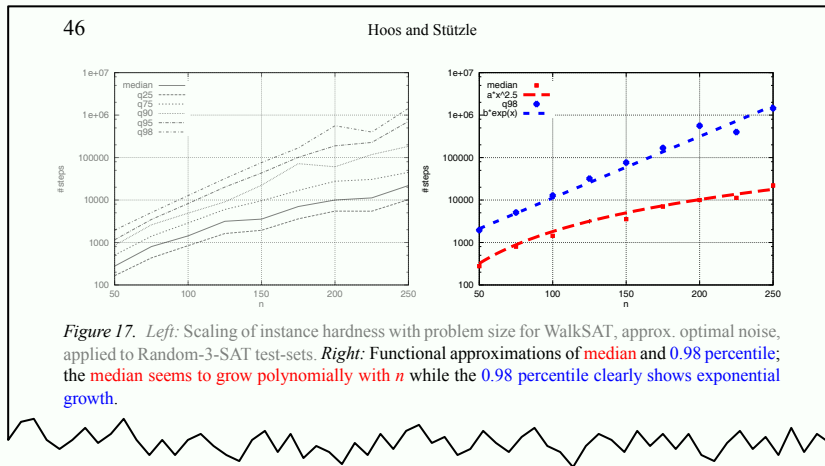
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(Taken from the paper “Local Search Algorithms for SAT: An Empirical Evaluation” by Hoos and Stützle, coloring added manually⁵⁹, copyright Springer.)

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(4)

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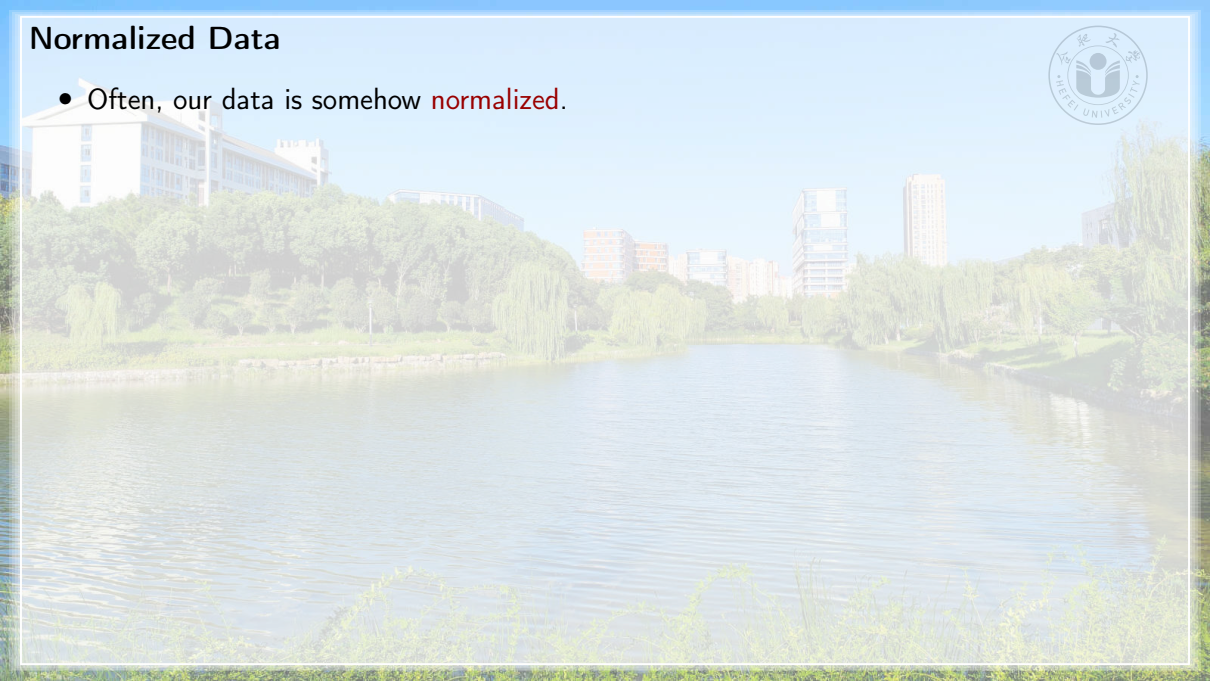
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I_1			
I_2			
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- We measure the required runtimes as follows:

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	
I_3	40 s	10 s	

Normalized Data



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Normalized Data



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- Let's say we solve the problem instances I_1 to I_3 with the different algorithms A_1 to A_3 .
- We measure the required runtimes as follows:

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s

Normalized Data



- Often, our data is somehow **normalized**.
- We measure the required runtimes as follows:
- The arithmetic mean values are the same.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s

Normalized Data



- Often, our data is somehow **normalized**.
- The arithmetic mean and the median values are the same.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s

Normalized Data



- Often, our data is somehow **normalized**.
- The arithmetic mean, the median, and the geometric mean values are the same.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

Normalized Data



- Often, our data is somehow **normalized**.
- The arithmetic mean, the median, and the geometric mean values are the same.
- We can conclude that the three algorithms offer the same performance in average over these benchmark instances.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

Normalized Data



- Often, our data is somehow **normalized**.
- We can conclude that the three algorithms offer the same performance in average over these benchmark instances.
- But often the measured numbers “look messier” and are harder to compare at first glance.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

Normalized Data



- Often, our data is somehow **normalized**.
- But often the measured numbers “look messier” and are harder to compare at first glance.
- So often we want to normalize them by picking one algorithm as “standard” and dividing them by its measurements.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

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- But often the measured numbers “look messier” and are harder to compare at first glance.
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- Let’s say A_1 was a well-known heuristic, maybe we even took its results from a paper, and we want to use it as baseline for comparison and normalize our data by it.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s



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median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s



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- Let's say A_1 was a well-known heuristic, maybe we even took its results from a paper, and we want to use it as baseline for comparison and normalize our data by it.
- OK, so we get this table with normalized values, which allow us to make sense of the data at first glance.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50

Normalized Data



- Often, our data is somehow **normalized**.
- OK, so we get this table with normalized values, which allow us to make sense of the data at first glance.
- If we now compute the arithmetic mean

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50
mean:	1.00	1.42	1.67



Normalized Data

- Often, our data is somehow **normalized**.
- OK, so we get this table with normalized values, which allow us to make sense of the data at first glance.
- If we now compute the arithmetic mean, then **A_1 is best**

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50
mean:	1.00	1.42	1.67



Normalized Data

- Often, our data is somehow **normalized**.
- OK, so we get this table with normalized values, which allow us to make sense of the data at first glance.
- If we now compute the arithmetic mean, then **A_1 is best** and **A_3 looks worst**.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50
mean:	1.00	1.42	1.67



Normalized Data

- Often, our data is somehow **normalized**.
- OK, so we get this table with normalized values, which allow us to make sense of the data at first glance.
- If we now compute the arithmetic mean, then **A_1 is best** and **A_3 looks worst**.
- According to the median

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50
mean:	1.00	1.42	1.67
median:	1.00	2.00	0.50



Normalized Data

- Often, our data is somehow **normalized**.
- If we now compute the arithmetic mean, then A_1 is **best** and A_3 looks **worst**.
- According to the median, A_3 is **best**

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50
mean:	1.00	1.42	1.67
median:	1.00	2.00	0.50



Normalized Data

- Often, our data is somehow **normalized**.
- If we now compute the arithmetic mean, then **A_1 is best** and **A_3 looks worst**.
- According to the median, **A_3 is best** and **A_2 is worst!**

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50
mean:	1.00	1.42	1.67
median:	1.00	2.00	0.50



Normalized Data

- Often, our data is somehow **normalized**.
- If we now compute the arithmetic mean, then A_1 is best and A_3 looks worst.
- According to the median, A_3 is best and A_2 is worst!
- Only the geometric mean still indicates that the algorithms perform the same...

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50
mean:	1.00	1.42	1.67
median:	1.00	2.00	0.50
geom:	1.00	1.00	1.00

Normalized Data



- Often, our data is somehow **normalized**.
- Hm.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50
mean:	1.00	1.42	1.67
median:	1.00	2.00	0.50
geom:	1.00	1.00	1.00

Normalized Data



- Often, our data is somehow **normalized**.
- Hm. OK, then let's normalize using the results of A_2 instead.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

Normalized Data



- Often, our data is somehow **normalized**.
- Hm. OK, then let's normalize using the results of A_2 instead.
- OK, so we get this table with normalized values.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	0.50	1.00	2.00
I_2	0.50	1.00	0.25
I_3	4.00	1.00	2.00

Normalized Data



- Often, our data is somehow **normalized**.
- OK, so we get this table with normalized values.
- If we now compute the arithmetic mean

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	0.50	1.00	2.00
I_2	0.50	1.00	0.25
I_3	4.00	1.00	2.00
mean:	1.67	1.00	1.42

Normalized Data



- Often, our data is somehow **normalized**.
- OK, so we get this table with normalized values.
- If we now compute the arithmetic mean, then **A_2 is best**

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	0.50	1.00	2.00
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Normalized Data



- Often, our data is somehow **normalized**.
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- If we now compute the arithmetic mean, then **A_2 is best** and **A_1 looks worst**.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
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- If we now compute the arithmetic mean, then **A_2 is best** and **A_1 looks worst**.
- According to the median

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Normalized Data

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- According to the median, A_1 is **best**

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- Often, our data is somehow **normalized**.
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- The geometric mean is the only meaningful average if we have **normalized** data!⁴³

Normalized Data



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- And we very often have normalized data.
- For example, at least half of the papers on the Job Shop Scheduling Problem (JSSP) normalize the result qualities they obtain on benchmark instances with the Best Known Solutions (BKSES) and then compute the arithmetic mean¹²¹.

Arithmetic Mean vs. Median vs. Geometric Mean

- Most publications report arithmetic mean results, many report median results, almost none report geometric means.



Arithmetic Mean vs. Median vs. Geometric Mean



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 - our algorithm has a bad worst-case behavior (which is also good to know).
 - If the median is much worse than the mean, then the mean is too optimistic, i.e., most of the time we should expect worse results.
- If there are outliers, the value of the arithmetic mean itself may be very different from any actually observed value, while the median is (almost always) similar to some actual measurements.

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- We therefore want to know both the arithmetic mean and the median.
- Often, our data is implicitly or explicitly normalized, e.g.,
 - if we divide result qualities by results of well-known heuristics or BKSes or
 - if we normalize the runtime using another algorithm as standard.

Arithmetic Mean vs. Median vs. Geometric Mean



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Arithmetic Mean vs. Median vs. Geometric Mean



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- I think: On raw data, compute all three measures of average, and pay special attention to the one looking the worst. On normalized data, compute the geometric mean, but also consider the arithmetic mean and median *if and only if they make **your** algorithm look worse.*

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- An average alone is not very meaningful – if we known nothing about the range of the data.
- We can therefore compute a measure of dispersion, i.e., a value that tells us whether the observations are stretched and spread far or squeezed tight around the center.

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The variance $\text{var}(A)$ of a data sample $A = (a_0, a_1, \dots, a_{n-1})$ with n observations can be estimated as:

$$\text{var}(A) = \frac{1}{n-1} \sum_{i=0}^{n-1} (a_i - \text{mean}(A))^2 = \frac{1}{n-1} \left[\left(\sum_{i=0}^{n-1} a_i^2 \right) - \frac{1}{n} \left(\sum_{i=0}^{n-1} a_i \right)^2 \right]$$

Standard Deviation



Definition: Sample Standard Deviation

The standard deviation $\text{sd}(A)$ of a data sample $A = (a_0, a_1, \dots, a_{n-1})$ with n observations is the square root of the estimated variance $\text{var}(A)$.

$$\text{sd}(A) = \sqrt{\text{var}(A)}$$

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- Large standard deviations indicate unreliable algorithms, but may also offer a potential that could be exploited: Given enough time, we can restart algorithms several times and expect to get different (and thus sometimes better) solutions.

Quantiles



Definition: Sample Quantile

The q -quantiles are the cut points that divide a sorted data sample $A = (a_0, a_1, \dots, a_{n-1})$ where $a_{i-1} \leq a_i \forall i \in 1 \dots (n-1)$ into q equally-sized parts.



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$$\begin{aligned} h &= (n-1) \frac{k}{q} \\ \text{quantile}_q^k(A) &= \begin{cases} a_h & \text{if } h \text{ is integer} \\ a_{\lfloor h \rfloor} + (h - \lfloor h \rfloor) * (a_{\lfloor h \rfloor + 1} - a_{\lfloor h \rfloor}) & \text{otherwise} \end{cases} \end{aligned}$$



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- Quantiles are a generalized form of the median.
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- 4-quantiles are called *quartiles*.
- We often consider *percentiles* or write things like “98% quantile” or “0.98 percentile” or “98% percentile” meaning $\text{quantile}_{100}^{98}$.

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- Two data samples A and B with $n_a = n_b = 19$ values.



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$$\text{mean}(A) = 7$$

$$B = (1, 3, 4, 4, 4, 5, 6, 6, 6, 6, 7, 7, 9, 9, 9, 10, 11, 12, 10'008)$$

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- Being based on the arithmetic mean, the variance and standard deviation are heavily influenced by outliers – with all pros and cons coming with that...

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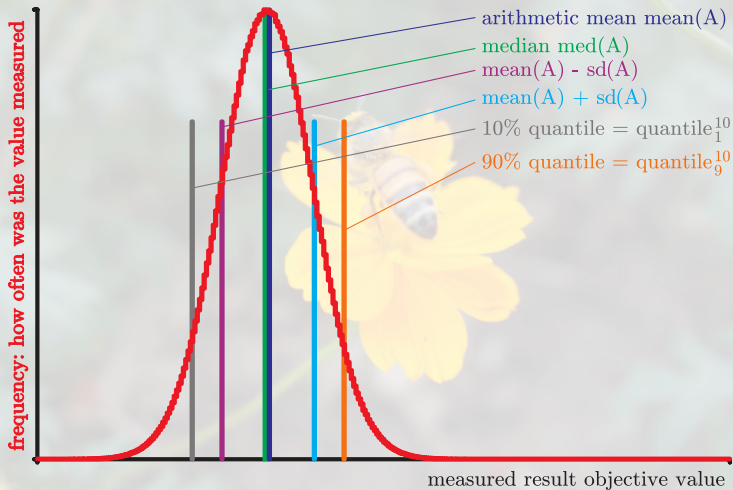
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- Being generalizations of the median, the quantiles are little influenced by outliers – with all pros and cons coming with that. . .

Further Example



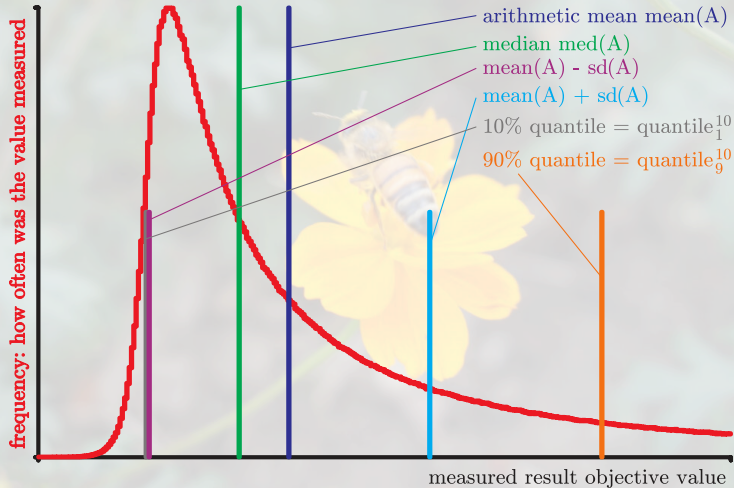
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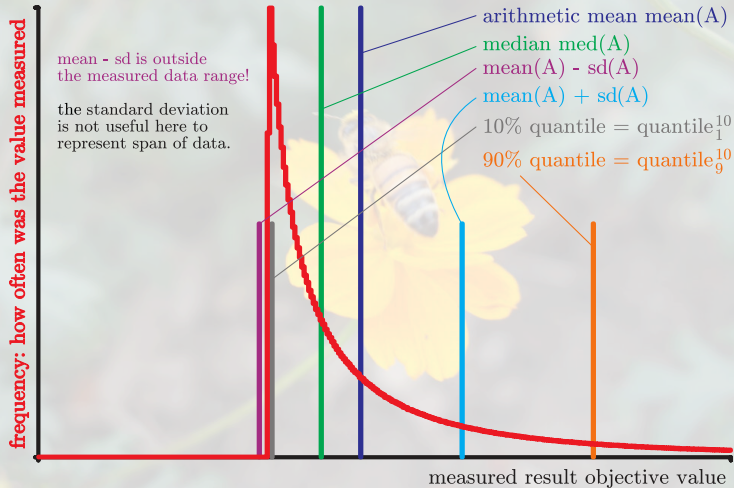
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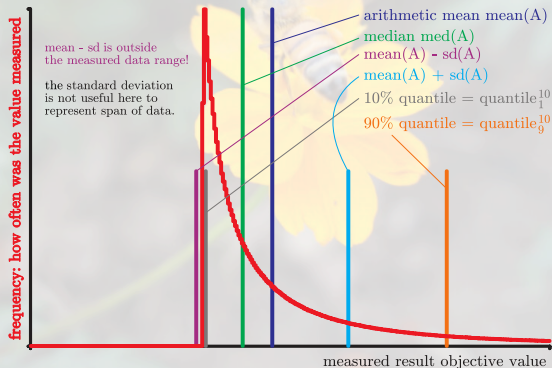
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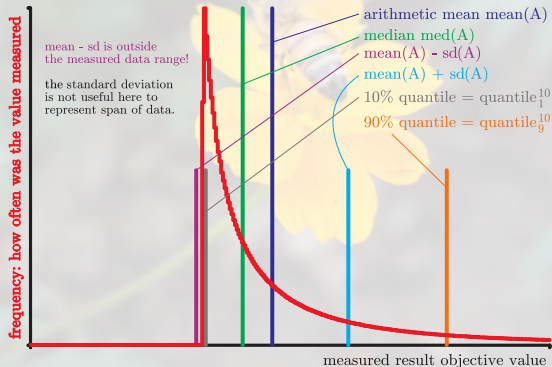
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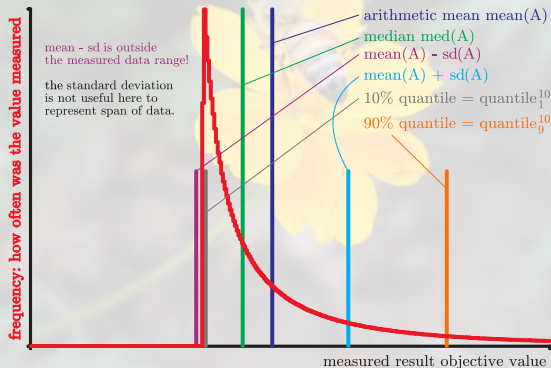
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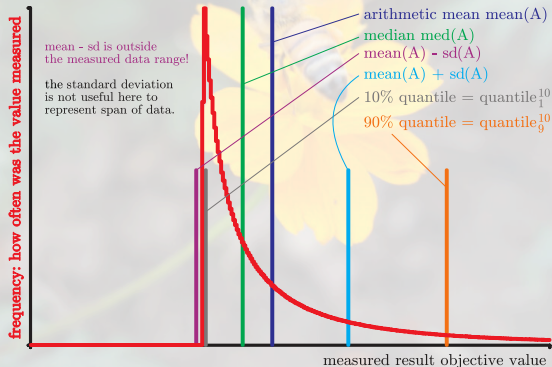
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 - A statement such as *“For this TSP instance, our algorithm can find tours with a length of $100 \pm 120 \text{ km}.$ ”* makes little sense. . .



Statistical Comparisons



Introduction



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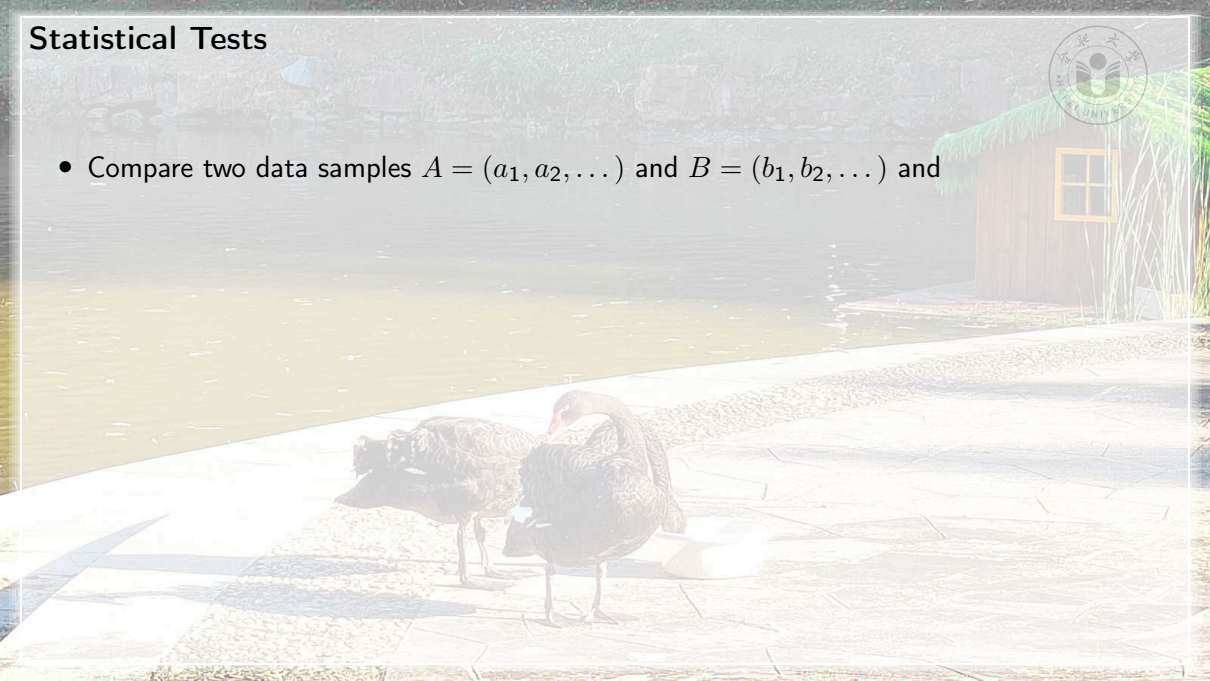


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- **The statement "*A is better than B*" makes only sense after we have decided about an upper bound α for the acceptable error probability p ! (and if $p < \alpha$, obviously)**

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- Disclaimer: I am not a mathematician. What follows are simplified explanations of concepts.

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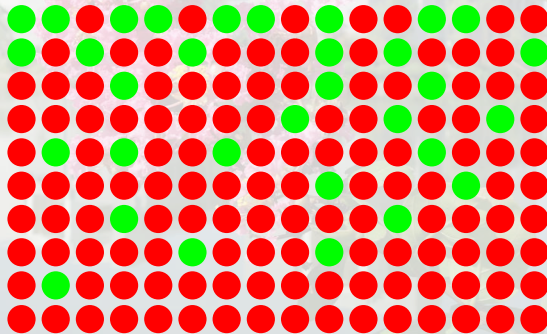
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- We play 160 times.
- I win 128 times. You win 32 times.
- Did I cheat? Is my coin “fixed?” (i.e., is your chance to win $\neq 0.5$)
- Assumption: I cheat. (alternative hypothesis H_1)
- It is impossible to compute my winning probability if I cheated...
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- (What we will do right now is called *binomial test*.)

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- The probability $P(k|n)$ to flip $k \in 0..n$ times heads (or tails) is thus:

$$P(k|n) = \binom{n}{k} 0.5^k * (1 - 0.5)^{n-k} = \binom{n}{k} 0.5^k * 0.5^{n-k} = \binom{n}{k} \frac{1}{2^n}$$

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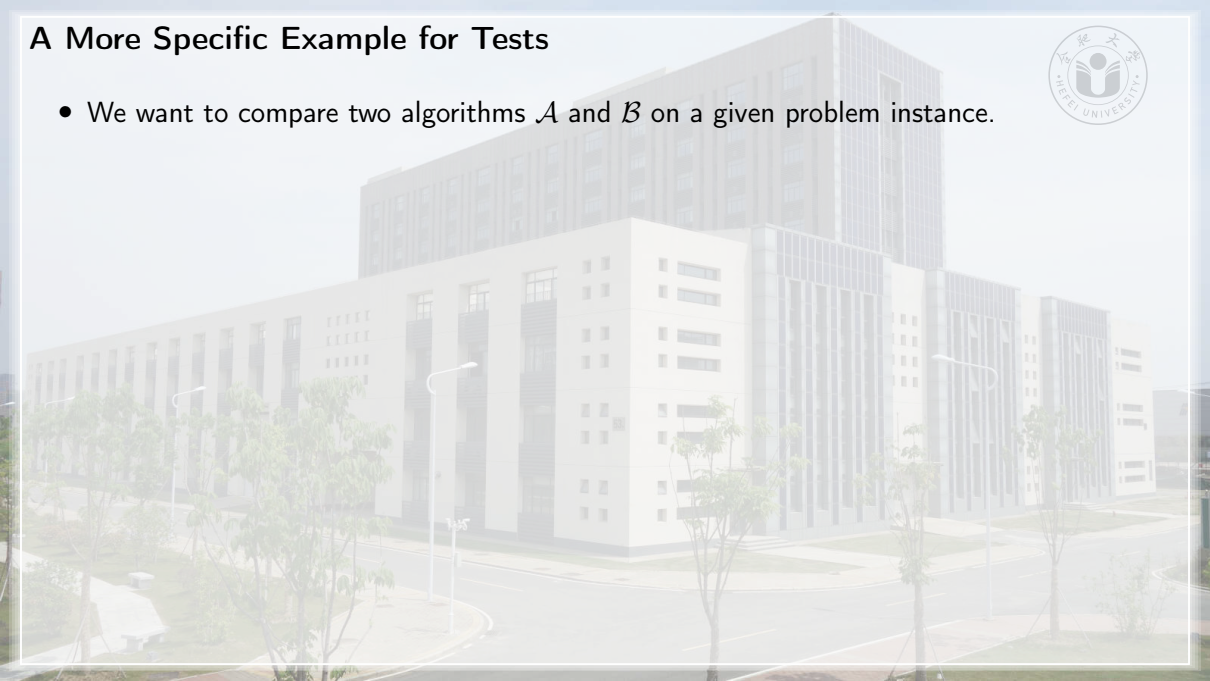


- Question: How likely is it that I win at least 128 times if I did not cheat?
- If the coin was an ideal coin, the chance that I win at least 128 out of 160 times is about $4 \cdot 10^{-15}$.
- If you claim that I cheat, your chance to be wrong is about $4 \cdot 10^{-15}$.
- Thus, if we cannot accept a chance p to be wrong higher than a significance level $\alpha = 1\%$, we can still say:

The observation is significant, I did likely cheat.

A More Specific Example for Tests

- We want to compare two algorithms $\mathcal{A}$ and $\mathcal{B}$ on a given problem instance.



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$$A = (2, 5, 6, 7, 9, 10)$$

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$$\text{mean}(A) = \frac{39}{6} = 6.5$$

$$\text{mean}(B) = \frac{16}{4} = 4$$

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$$\begin{aligned}\text{mean}(A) &= \frac{39}{6} = 6.5 \\ \text{mean}(B) &= \frac{16}{4} = 4\end{aligned}$$

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- But is this assumption justified based on the data we have?

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- It looks like algorithm B may produce the smaller objective values.
- But is this assumption justified based on the data we have?
- Is the difference between $\text{mean}(A)$ and $\text{mean}(B)$ significant at a threshold of $\alpha = 2\%$?

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- If $\mathcal{B}$ is truly better than $\mathcal{A}$, which is our hypothesis H_1 , then we cannot calculate anything.

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- Let's use a Python¹²³ program to test the combinations

A More Specific Example



```
1  """Enumerate all combinations of numbers 1 to 10."""
2  mean_lower_or_equal_to_4 = 0  # how often did we find a mean <= 4
3  total_combinations      = 0  # total number of tested combinations
4
5  for i in range(1, 11):        # i goes from 1 to 10
6      for j in range(1, i):     # j goes from 1 to i - 1
7          for k in range(1, j): # k goes from 1 to j - 1
8              for l in range(1, k): # l goes from 1 to k - 1
9                  if ((i + j + k + l) / 4) <= 4: # check for extreme case
10                     mean_lower_or_equal_to_4 += 1 # count extreme case
11                     total_combinations += 1      # count all combinations
12
13  print(f" combinations with mean <= 4: {mean_lower_or_equal_to_4}")
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```

```
1  combinations with mean <= 4: 27
2  total number of combinations: 210
```

A More Specific Example



$$O = A \cup B = (1, 2, 3, 4, 5, 6, 7, 8, 9, 10)$$

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- There are $\binom{10}{4} = 210$ different ways to draw 4 (or 6) elements from O
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- There are 27 such combinations with a mean of **less or equal 4**.

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- The probability p to observe a situation at least as extreme as A and B under H_0 is thus:

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$$p = \frac{\text{\#cases } C : \text{mean}(C) \leq \text{mean}(B)}{\text{\#all cases}} = \frac{27}{210} = \frac{9}{70} \approx 0.1286$$

A More Specific Example



- Extreme cases into the other direction are the same, because if $\text{mean}(B) \leq 4$ then $\text{mean}(A) \geq 6.5$ for any division $A \cup B = O$ and vice versa.

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$$\sum_{\forall o \in O} o = \sum_{o=1}^{10} o = \frac{10(10+1)}{2} = 55$$

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$$\text{mean}(B) = \left(\frac{1}{4} \sum_{\forall b \in B} b \right) \leq 4 \implies \left(\sum_{\forall b \in B} b \right) \leq 4 * 4 \leq 16$$

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- So – of course – we could have also done the test the other way around with the same result!

A More Specific Example



- The probability p to observe a constellation at least as extreme as A or B under H_0 is thus:

$$p = \frac{\text{\#cases } C : \text{mean}(C) \leq \text{mean}(B)}{\text{\#all cases}} = \frac{27}{210} = \frac{9}{70} \approx 0.1286$$

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- If we claim that A and B are from distributions with means as different as observed. . .

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- **The method here is only feasible for small sample sets, real tests are more sophisticated**

Statistical Tests: Types

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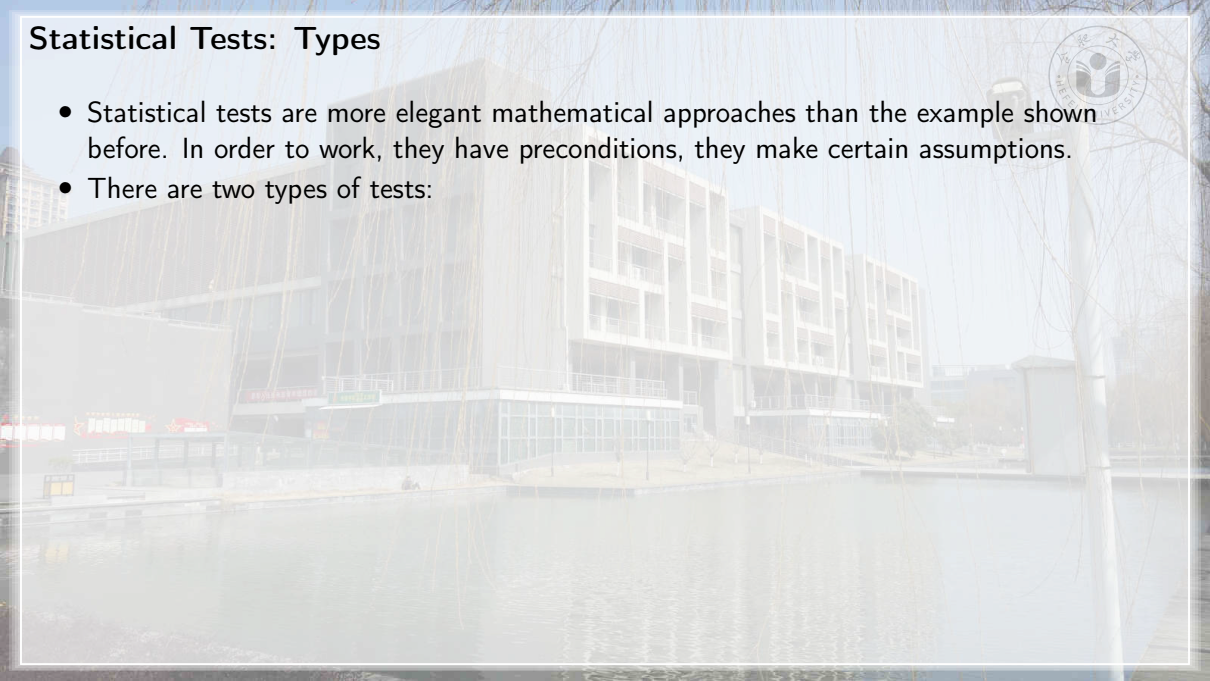
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 - **Often, the most suitable test is the Mann-Whitney U test.**

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- To be **practically significant**, the measured difference of results should be **large enough** and statistically significant already with few runs, say, 11 or 21, not just with ≥ 100 runs.

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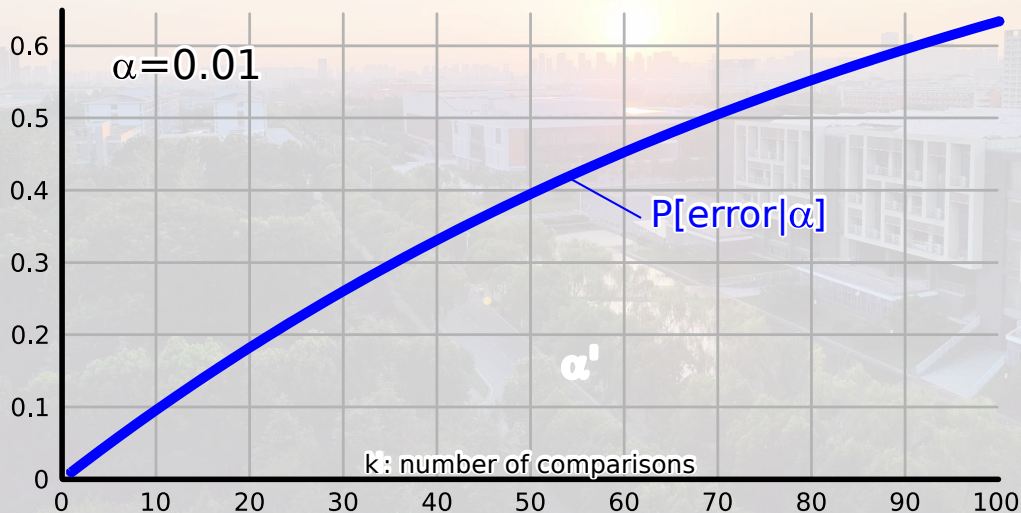
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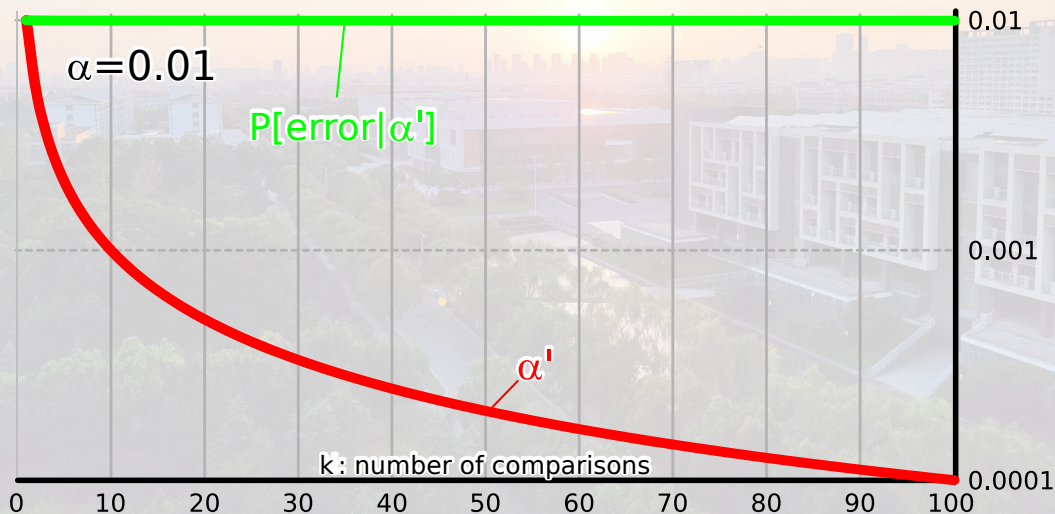
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- The Bonferroni correction is very conservative and not suitable when checking independent parameters⁹⁶, but for comparing optimization algorithm, it does the trick. Being conservative is good.



Testing is Not Enough



The question of termination



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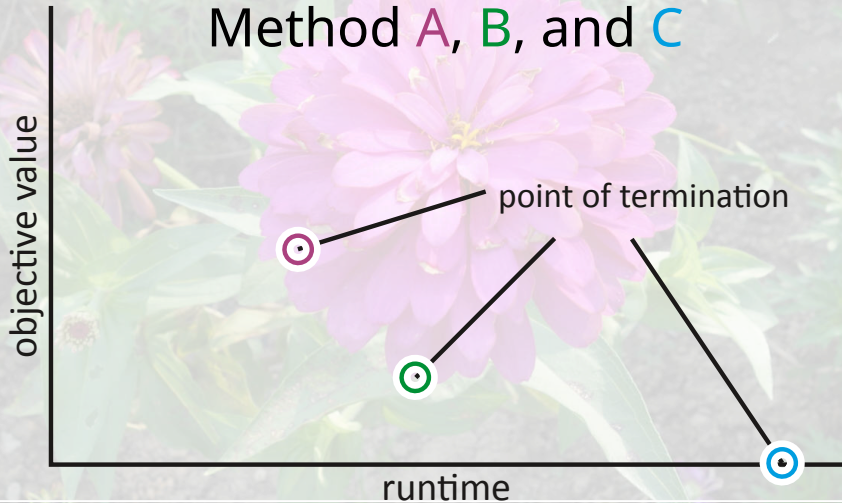


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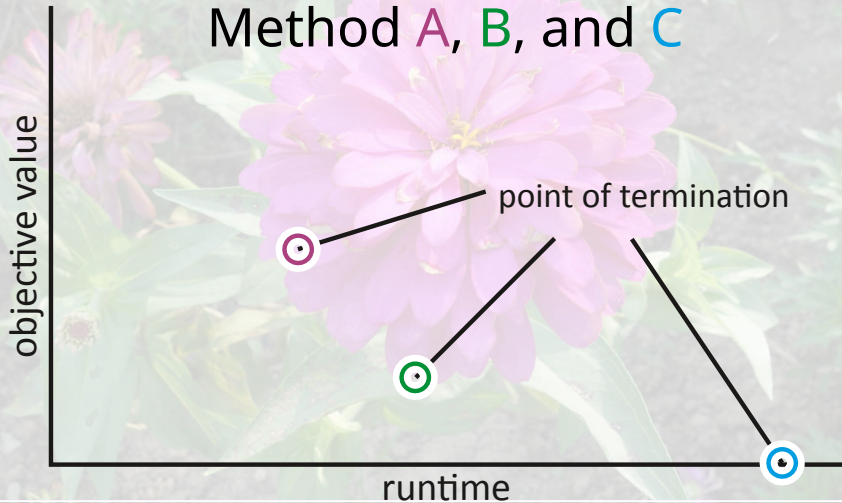


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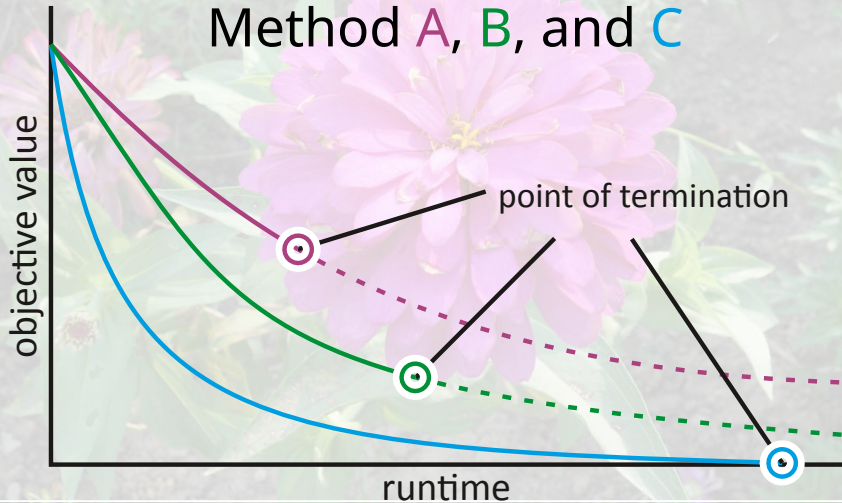
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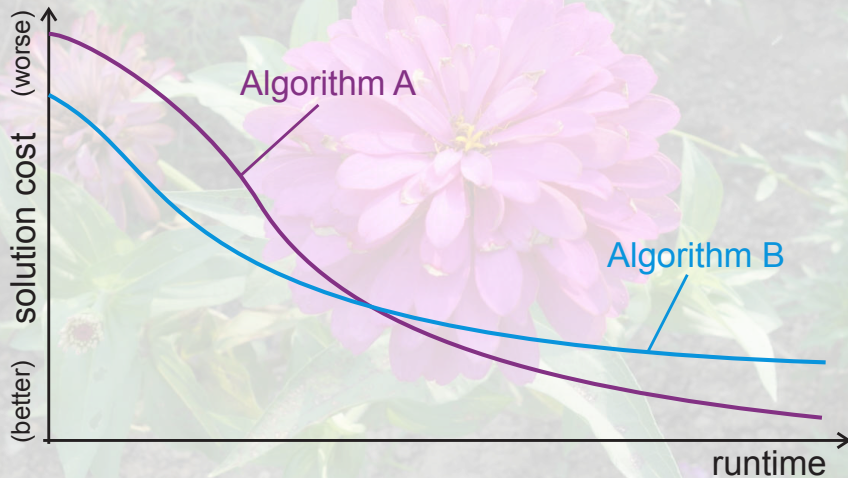
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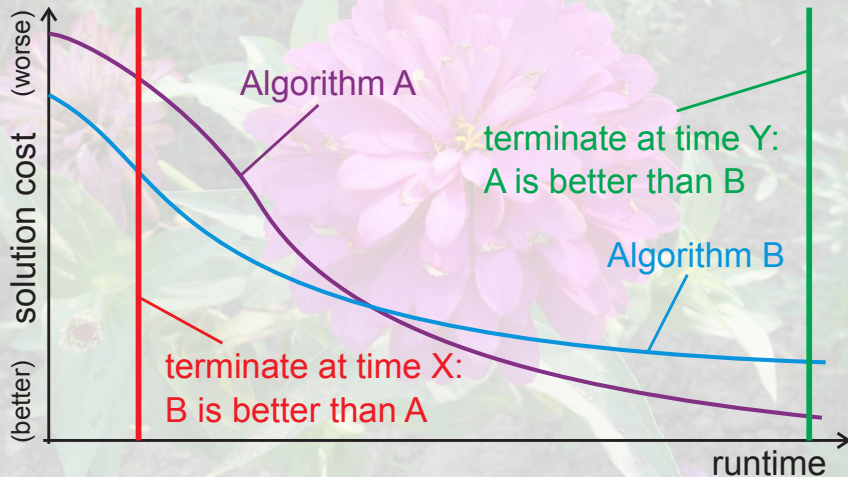
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Other Stuff



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 - Only the benchmark instances where the algorithm performs well are chosen. Be wary of statements such as “We now present the results of our algorithm on 10 of the TSPLib instances.” (TSPLib has more than 100. . .)
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Cheating



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- Sometimes, results may be straight up fabricated.
- Misleading statistics are reported
- Uneven configuration effort: Much effort is spent on configuring the own algorithm, the algorithms used for comparison are used with bad settings.

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- Misleading statistics are reported
- Uneven configuration effort.
- Incomparable results are reported (see our discussion on why testing is not enough).

Cheating



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Reproducibility prevents cheating and misunderstandings!



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- Use well-known benchmarks, provide your source code!



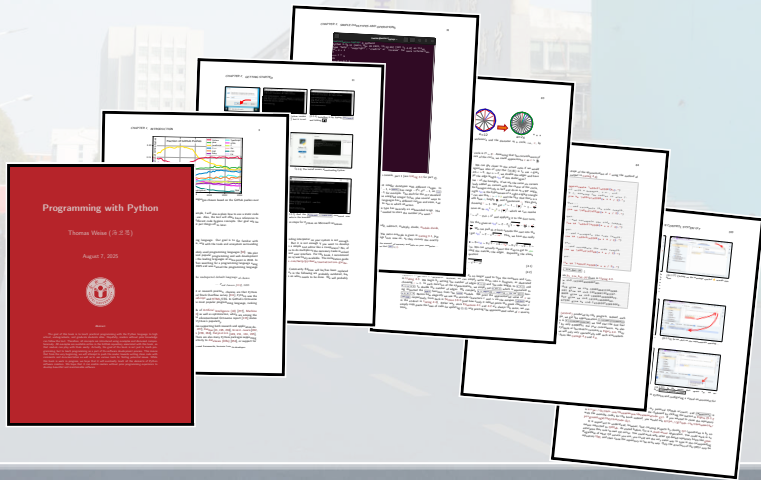
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Programming with Python



We have a freely available course book on *Programming with Python* at <https://thomasweise.github.io/programmingWithPython>, with focus on practical software development using the Python ecosystem of tools¹²³.



Databases



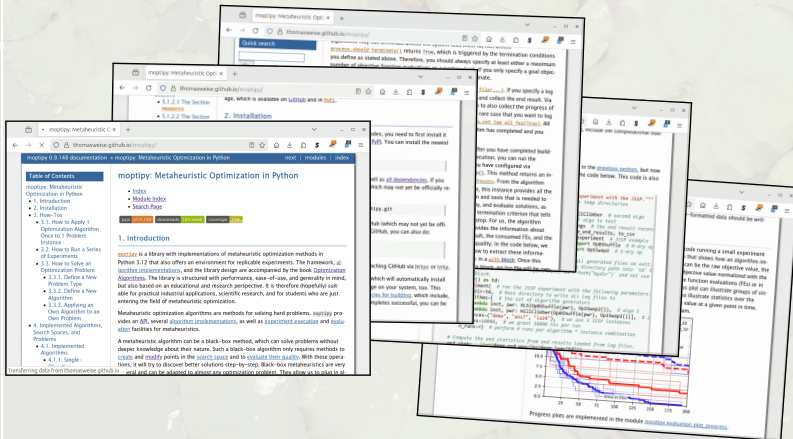
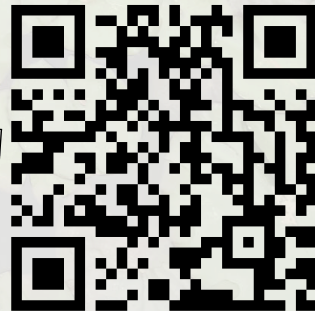
We have a freely available course book on *Databases* at <https://thomasweise.github.io/databases>, with actual practical examples using a real database management system (DBMS)¹¹⁹.



Metaheuristic Optimization in Python: moptipy



We offer moptipy¹²⁸, a mature open source Python package for metaheuristic optimization, which implements several algorithms, can run self-documenting experiments in parallel and in a distributed fashion, and offers statistical evaluation tools.

The image shows a collage of screenshots from the moptipy GitHub repository and its documentation. The top left shows the GitHub repository page with the README. The top right shows the documentation page with a 'Table of Contents' and '1. Introduction'. The bottom left shows a 'Table of Contents' for the documentation. The bottom right shows a 'Progress plots' graph with multiple lines representing different optimization runs. The graph has a y-axis from 0.0 to 10.0 and an x-axis from 0 to 200. The lines show a general downward trend, indicating improvement over time.



谢谢你们！
Thank you!
Vielen Dank!



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Glossary I



- (1 + 1) EA** The (1 + 1) EA is a local search algorithm that retains the best solution x_c discovered so far during the search^{20,37}. In each step, it applies a unary search operator to this best-so-far solution x_c and derives a new solution x_n . If the new solution x_n is *better or equally good* when compared with x_c , i.e., not worse, then it replaces it, i.e., is stored as the new x_c . If the search space are bit strings of length n , then the (1 + 1) EA uses a unary search operator that flips each bit independently with probability m/n , where usually $m = 1$. This operator is the main difference to randomized local search (RLS). The (1 + 1) EA is a special case of the $(\mu + \lambda)$ evolutionary algorithm $((\mu + \lambda)$ EA) where $\mu = \lambda = 1$.
- (1 + 1) FEA** The (1 + 1) evolutionary algorithm $((1 + 1)$ EA) with Frequency Fitness Assignment (FFA) plugged in^{129,130,137}. On the discrete bit-string domains, this algorithm has a very specific structure: Its unary search operator flips each bit with probability $1/n$, where n is the dimension of the search space, distinguishing it from the unary operator in RLS with FFA (FRLS) which flips exactly one bit. On other domains, such as permutations, the term is used synonymously with FRLS.
- $(\mu + \lambda)$ EA** The $(\mu + \lambda)$ EA is an evolutionary algorithm (EA) where, in each generation, λ offspring solutions are generated from the current population of μ parent solutions. The offspring and parent populations are merged, yielding $\mu + \lambda$ solutions, from which then the best μ solutions are retained to form the parent population of the next generation. If the search space is the bit strings of length n , then this algorithm usually applies a mutation operator flipping each bit independently with probability $1/n$.
- ACO** *Ant Colony Optimization* is a nature-inspired optimization method for combinatorial problems where solutions are generated by “ants” that move from node to node in a graph choosing edges based on (1) the simulated pheromone on the edges and (2) a per-edge heuristic value^{34–36}. If an ant produced a good solution, “pheromone” is distributed over the edges it visited, making it more likely to be re-visited by other ants.
- BKS** The *Best Known Solution* for an instance of an optimization problem is the best solution (measured based on the objective values) that has ever been reported in literature. BKSes are not necessarily globally optimal, as in many instances of $\mathcal{NP}$ -hard problems, the true optima are unknown.
- C** is a programming language, which is very successful in system programming situations^{32,100,122}.
- Concorde** is an exact solver for the TSP^{3,27,28}, which could even solve the largest instance of TSPLib (with 85 900 cities) to optimality.

Glossary II



- CSV** *Comma-Separated Values* is a very common and simple text format for exchanging tabular or matrix data ¹⁰⁹. Each row in the text file represents one row in the table or matrix. The elements in the row are separated by a fixed delimiter, usually a comma (","), sometimes a semicolon (";"). Python offers some out-of-the-box CSV support in the `CSV` module ²⁹.
- DB** A *database* is an organized collection of structured information or data, typically stored electronically in a computer system. Databases are discussed in our book *Databases* ¹¹⁰.
- DBMS** A *database management system* is the software layer located between the user or application and the database (DB). The DBMS allows the user/application to create, read, write, update, delete, and otherwise manipulate the data in the DB ¹³⁵.
- EA** An *evolutionary algorithm* is a metaheuristic optimization method that maintains a population of candidate solutions, which undergo selection (where better solutions are chosen with higher probability) and reproduction (where mutation and recombination create a new candidate solution from one or two existing ones, respectively) ^{6,120}.
- FE** *Objective function evaluations* are an implementation-independent measure of runtime for optimization algorithms ^{7,124,125,128}. 1 FE equals to one evaluated candidate solution during the optimization process.
- FFA** *Frequency Fitness Assignment* (中文: 频率适应度分配) is a algorithm plugin for optimization methods applied to discrete or combinatorial problems with not-too-many different possible objective values ^{126,129,130}. It replaces the objective values in all comparisons with their absolute encounter frequency so far during the search. FFA has successfully been applied to the MaxSAT ^{129,130}, the TSP ^{75,76}, and the Quadratic Assignment Problem (QAP) ^{24,116}, amongst others. Its properties have been intensely researched on discrete benchmark functions such as OneMax, LeadingOnes, Trap, TwoMax, Jump, and Plateau ^{129,130,137}. The most common realizations of FFA so far are the (1 + 1) FEA and FRLS.
- Git** is a distributed Version Control Systems (VCS) which allows multiple users to work on the same code while preserving the history of the code changes ^{111,117}. Learn more at <https://git-scm.com>.
- GitHub** is a website where software projects can be hosted and managed via the Git VCS ^{95,117}. Learn more at <https://github.com>.

Glossary III



Java is another very successful programming language, with roots in the C family of languages ^{14,79}.

JSSP The *Job Shop Scheduling Problem* ^{13,25,72} is one of the most prominent and well-studied scheduling tasks. In a JSSP instance, there are k machines and m jobs ¹¹². Each job must be processed once by each machine in a job-specific sequence and has a job-specific processing time on each machine. The goal is to find an assignment of jobs to machines that results in an overall shortest makespan, i.e., the schedule which can complete all the jobs in the shortest time. The JSSP is $\mathcal{NP}$ -complete ^{23,72}.

Jump The Jump problem is a benchmark problem for bit-string based black-box search ^{37,45,137}. Like the OneMax, it tries to maximize the number of 1 bits in a bit string of chosen length n . However, the solutions representing the last ω bits around the optimum to flip are deceptive, meaning that they are actually worse and getting worse than surrounding solutions. The objective function $f(x)$ of the minimization variant is $n - \|x\|_1$ if $\|x\|_1 = n$ or $\|x\|_1 \leq n - \omega$ hold, and $\omega + \|x\|_1$ otherwise, where $\|x\|_1 = \sum_{i=1}^n x_i$. This means that, the larger ω , the larger the deceptive basin around the optimum that an algorithm needs to cross. An $(1 + 1)$ EA will converge to the resulting local optimum and thus have an expected runtime of $\Theta(n^\omega + n \ln n)$ ³⁷. In experiments, the $(1 + 1)$ EA with FFA ($(1 + 1)$ FEA) achieves polynomial runtime on Jump, seemingly in $\mathcal{O}(n^2 \ln n)$ ^{129,130,137}.

LeadingOnes The LeadingOnes problem is a benchmark problem for bit-string based black-box search ^{105,132,137}. It tries to maximize the number of *leading* 1 bits in the bit string of a chosen dimension n . The objective function $f(x)$ of the minimization variant of this problem can be defined as $f(x) = n - \sum_{i=1}^n \prod_{j=1}^i x_j$. Different from OneMax, the contribution to the objective value of the bit at index i thus depends on the state of all the bits *before* this index. The problem is unimodal, i.e., has only a single global optimum and no local optimum. Different from OneMax, it is not separable, i.e., we can optimize each variable separately. The expected runtime of $(1 + 1)$ EA on LeadingOnes is in $\Theta(n^2)$ ³⁷. The average runtime $(1 + 1)$ FEA on LeadingOnes seems to be slower by factor proportional to n , i.e., seems to be in $\mathcal{O}(n^3)$ ¹²⁹.

Glossary IV



- MaxSAT** The goal of satisfiability problems is to find an assignment for n Boolean variables that make a given Boolean formula $F : \{0, 1\}^n \mapsto \{0, 1\}$ become true. In the *Maximum Satisfiability* problem (MaxSAT)⁶⁰, F is given in conjunctive normal form, i.e., the variables appear either directly or negated in m “or” clauses, which are all combined into one “and.” The objective function $f(x)$, subject to minimization, computes the number of clauses which are false under the variable setting x . If $f(x) = 0$, then all clauses of F are true, which solves the problem. The MaxSAT problem is $\mathcal{NP}$ -complete²⁶ and exhibits low epistasis but deceptiveness¹⁰², making it hard for local searches.
- metaheuristic** A *metaheuristic* is an optimization algorithm which performs a cycle of trial and error during which candidate solutions are sampled from the solution space $\mathbb{Y}$ and evaluated using an objective function $f : \mathbb{Y} \mapsto \mathbb{R}$ ^{49,120}. The results of the objective function are used to decide which solution is promising. The neighborhoods around promising solutions or similarly-structured solutions are then sampled in the next cycle. Prominent metaheuristics are stochastic local searches (SLSes), Simulated Annealing (SA), TS, and EAs.
- moptipy** is the *Metaheuristic Optimization in Python* library¹²⁸. It has been used in several different research works, including^{24,75–77,116,134,138,139}. Algorithms implemented by moptipy include EAs, $(1 + 1)$ EA, RLS, SA, and the $(1 + 1)$ FEA, amongst others. Learn more at <https://thomasweise.github.io/moptipy> and <https://thomasweise.github.io/moptipyapps>.
- OneMax** OneMax is the simplest possible benchmark problem for bit-string based black-box search^{38,88,137}. The goal is to maximize the number of 1 bits in the bit string of a chosen dimension n . The objective function $f(x)$ of the minimization variant of this problem can be defined as $f(x) = n - \sum_{i=1}^n x_i$. The problem is unimodal, i.e., has only a single global optimum and no local optimum. It is fully separable, i.e., we can optimize each variable separately. The expected runtime of $(1 + 1)$ EA on OneMax is in $\Theta(n \ln n)$ ^{33,88}. The average runtime $(1 + 1)$ FEA on OneMax seems to be slower by factor proportional to n , i.e., seems to be in $\mathcal{O}(n^2 \ln n)$ ¹²⁹.

Glossary V



- Plateau** The Plateau problem is a benchmark problem for bit-string based black-box search^{2,137}. It is structured exactly like Jump, but instead of being deceptive, the last ω bits to flip around the optimum are neutral. The objective function $f(x)$ of the minimization variant is $n - \|x\|_1$ if $\|x\|_1 = n$ or $\|x\|_1 \leq n - \omega$ hold, and ω otherwise, where $\|x\|_1 = \sum_{i=1}^n x_i$. This means that, the larger ω , the larger the neutral plane around the optimum that an algorithm needs to cross. An $(1 + 1)$ EA has an expected runtime of $\Theta(n^\omega)$ on plateaus². The average runtime of the $(1 + 1)$ FEA on Plateaus is slower by a factor seemingly proportional in n .
- PostgreSQL** An open source object-relational DBMS^{41,90,98,114}. See <https://postgresql.org> for more information.
- Python** The Python programming language^{61,74,81,123}, i.e., what you will learn about in our book¹²³. Learn more at <https://python.org>.
- QAP** The *Quadratic Assignment Problem* is an optimization problem where the goal is to assign a set of n facilities to a set of n locations^{18,24,70,78}. Such an assignment can be represented as a permutation x of the first n natural numbers, where x_i specifies the location where facility i should be placed. For each QAP, a distance matrix D is given, where D_{pq} specifies the distance from location p to location q , as well as a flow matrix F , where F_{ij} is the amount of material flowing from facility i to facility j . The objective function $f(x)$ then rates a permutation x as $f(x) = \sum_{i=1}^n \sum_{j=1}^n D_{x_i x_j} F_{ij}$. The QAP is $\mathcal{NP}$ -complete¹⁰⁷.
- RLS** Randomized local search retains the best solution x_c discovered so far during the search and, in each step, it applies a unary search operator to this best-so-far solution x_c and derives a new solution x_n . If the new solution x_n is *better or equally good* when compared with x_c , i.e., not worse, then it replaces it, i.e., is stored as the new x_c . If the search space are bit strings of length n , then RLS uses a unary search operator that flips exactly one bit. This operator is the main difference to $(1 + 1)$ EA.
- FRLS** The RLS with FFA plugged in¹³⁷. On the discrete bit-string domains, this algorithm has a very specific structure: Its unary search operator flips exactly one bit, distinguishing it from the unary operator in the $(1 + 1)$ FEA, which flips each bit with probability $1/n$. On other domains, such as permutations, we use the term synonymously with $(1 + 1)$ FEA.
- SA** Simulated Annealing is a SLS that sometimes accepts a worse solution^{21,64,67,97}. The probability to do so decreases over time and with the difference in objective values, i.e., is the lower the worse the new solution is.

Glossary VI



- SGA** The Standard Genetic Algorithm ^{6,30,51,85,87,120} was the first EA. It maintains a population of solutions and applies mutation and crossover to generate offspring solutions. It uses fitness proportionate selection to choose which solutions should “survive” into the next generation, which today is considered a very bad design choice, see, e.g., ¹³².
- SLS** *Stochastic local search* or *local search* for short, is one of the simplest metaheuristic search paradigms ⁶⁰: A single “current” candidate solution is maintained and new solutions are sampled from its neighborhood. If these solutions are deemed interesting based on their objective value, they replace the “current”. The cycle continues. Basic examples are RLS and $(1 + 1)$ EA. More complex examples that add mechanisms to escape local optima on top of the SLS paradigm are TS and SA.
- SOTA** State-of-the-Art, i.e., the best-performing methods for a field or problem type
- Trap** The Trap problem is a benchmark problem for bit-string based black-box search ^{37,89,137}. It is structured exactly like OneMax, however the worst and the best solution are swapped. This makes the problem highly deceptive, as following the “gradient” in the search space leads the search away from the global optimum. For bit strings of length n , the objective function $f(x)$ of the minimization variant is 0 if $\|x\|_1 = 0$ and $n - \|x\|_1 - 1$ otherwise, where $\|x\|_1 = \sum_{i=1}^n x_i$. An $(1 + 1)$ EA will converge to the resulting local optimum and thus have an expected runtime of $\Theta(n^n)$ ³⁷. In experiments, the $(1 + 1)$ FEA achieves polynomial runtime on Trap, seemingly in $\mathcal{O}(n^2 \ln n)$ ^{129,130,137}.
- TS** *Tabu Search* is one of the most successful SLSes ^{47,48,50}. In order to avoid returning to previous solutions, it declares certain solutions or search moves as “tabu.” For example, if a bit in a discrete problem has been flipped by the search operator, it may be forbidden to flip the bit again for some time, thus preventing the algorithm from getting trapped in a cycle. This rule may be overridden if a new best-so-far solution is discovered that outperforms all previous solutions, in which case it would be accepted (this is the so-called aspiration criterion).
- TSP** In an instance of the *Traveling Salesperson Problem*, also known as *Traveling Salesman Problem*, a set of n cities or locations as well as the distances between them are defined ^{3,54,73,75,124,131}. The goal is to find the shortest round-trip tour that starts at one city, visits all the other cities one time each, and returns to the origin. The TSP is one of the most well-known $\mathcal{NP}$ -hard combinatorial optimization problems ^{46,54}.

Glossary VII



- TSPLib** is a library of benchmark instances for the Traveling Salesperson Problem (TSP) ^{103,104}. Its original URL was available until 2025 and then became defunct. The repository <https://github.com/shredderzsj/TSPLIB> contains the data. Additionally, you may also check <https://github.com/thomasWeise/tspSuite>, which contains all the TSPLib instances as well as Java code to load them. The TSPLib contains 110 instances of symmetric TSPs with scales n ranging from 14 to 85 900, but 93 instances have less than 2000 nodes. There also are 19 asymmetric TSPs with scales $n \in 17..443$.
- TwoMax** The TwoMax problem is a benchmark problem for bit-string based black-box search ^{45,118,137}. It has a global and a local optimum, both of about the same size, at opposite ends of the search space. For bit strings of length n , the objective function $f(x)$ of the minimization variant is 0 if $\|x\|_1 = n$ and $1 + n - \max\{\|x\|_1, n - \|x\|_1\}$ otherwise, where $\|x\|_1 = \sum_{i=1}^n x_i$. An $(1 + 1)$ EA has a 50/50 chance to converge to the local or global optimum and thus has an expected runtime in $\Theta(n^n)$ ^{44,45}. In experiments, the $(1 + 1)$ FEA achieves polynomial runtime seemingly in $\mathcal{O}(n^2 \ln n)$ on TwoMax ^{129,130,137}.
- unit test** Software development is centered around creating the program code of an application, library, or otherwise useful system. A *unit test* is an *additional* code fragment that is not part of that productive code. It exists to execute (a part of) the productive code in a certain scenario (e.g., with specific parameters), to observe the behavior of that code, and to compare whether this behavior meets the specification ^{9,91–93,106,115}. If not, the unit test fails. The use of unit tests is at least threefold: First, they help us to detect errors in the code. Second, program code is usually not developed only once and, from then on, used without change indefinitely. Instead, programs are often updated, improved, extended, and maintained over a long time. Unit tests can help us to detect whether such changes in the program code, maybe after years, violate the specification or, maybe, cause another, depending, module of the program to violate its specification. Third, they are part of the documentation or even specification of a program.
- URI** A *Uniform Resource Identifier* is an identifier for an abstract or physical resource in the internet ¹³⁶. It can be a URL, a name, or both. URIs are supersets of URLs. The connection strings of the PostgreSQL DBMS are examples for URIs.
- URL** A *Uniform Resource Locator* identifies a resource in the World Wide Web (WWW) and a way to obtain it by describing a network access mechanism. The most notable example of URLs is the text you write into web browsers to visit websites ¹¹. URLs are subsets of Uniform Resource Identifiers (URIs).

Glossary VIII



VCS A *Version Control System* is a software which allows you to manage and preserve the historical development of your program code¹¹⁷. A distributed VCS allows multiple users to work on the same code and upload their changes to the server, which then preserves the change history. The most popular distributed VCS is Git.

WWW World Wide Web^{10,31}

$i!$ The factorial $a!$ of a natural number $a \in \mathbb{N}_1$ is the product of all positive natural numbers less than or equal to a , i.e., $a! = 1 * 2 * 3 * 4 * \dots * (a - 1) * a$ ^{22,39,80}.

$i..j$ with $i, j \in \mathbb{Z}$ and $i \leq j$ is the set that contains all integer numbers in the inclusive range from i to j . For example, $5..9$ is equivalent to $\{5, 6, 7, 8, 9\}$

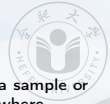
$f(y)$ The *objective function* $f : \mathbb{Y} \mapsto \mathbb{R}$ can compute a cost or rating for each candidate solution y in the solution space $\mathbb{Y}$. The smaller this value, the better the solution, i.e., in the context of this work, objective functions are subject to minimization. Objective functions are not necessarily continuous and it is also quite common that they have integer or natural numbers as results, e.g., we often have $f : \mathbb{Y} \mapsto \mathbb{N}_0$ instead of $f : \mathbb{Y} \mapsto \mathbb{R}$.

$\gamma(x)$ The decoding function $\gamma : \mathbb{X} \mapsto \mathbb{Y}$ translates point from the search space $\mathbb{X}$ to one point y in the solution space $\mathbb{Y}$. It is a very common case that $\mathbb{X} = \mathbb{Y}$ and that the decoding function γ is the identity mapping.

geom(A) The *geometric mean* $\text{geom}(A)$ is the n^{th} root of the product of n positive values in a dataset $A = (a_0, a_1, \dots, a_{n-1})$ with $a_i > 0$ for all $i \in 0..n$, i.e., $\text{geom}(A) = \sqrt[n]{\prod_{i=0}^{n-1} a_i} = \exp\left(\frac{1}{n} \sum_{i=0}^{n-1} \log a_i\right)$.

mean(A) The *arithmetic mean* $\text{mean}(A)$ is an estimate of the expected value of a distribution from which a data sample was, well, sampled. Its is computed on data sample $A = (a_0, a_1, \dots, a_{n-1})$ as the sum of all n elements a_i in the sample data A divided by the total number n of values, i.e., $\text{mean}(A) = \frac{1}{n} \sum_{i=0}^{n-1} a_i$.

Glossary IX



median(A) The *median* median(A) is the value separating the bigger-valued half from the smaller-valued half of a data sample or distribution. Its estimate is the value right in the middle of a *sorted* data sample $A = (a_0, a_1, \dots, a_{n-1})$ where $a_{i-1} \leq a_i \forall i \in 1 \dots (n-1)$ with an odd number of elements and the mean of the two values in the middle if n is even. In other words, $\text{median}(A) = a_{\frac{n-1}{2}}$ if n is odd and $\frac{1}{2} \left(a_{\frac{n}{2}-1} + a_{\frac{n}{2}} \right)$ otherwise, i.e., if n is even.

n We usually use n to denote the scale, i.e., the size of the problem to solve. For example, in a TSP, this may be the number of cities, in a discrete benchmark problem such as OneMax or Trap, this may be the number of bits over which the problem is defined, in a QAP, this may be the number of facilities and locations, and in a MaxSAT problem, this may be the number of variables that are input to the Boolean formula that we want to become true. There are also problems where the scale is not defined straightforwardly like this. For example, in a JSSP, we have a number of machines and a number of jobs, which together define the size of the problem. But even then, the scale n relates to the size of the solution space $\mathbb{Y}$, and, hence, also of the search space $\mathbb{X}$, which usually grow exponentially with n . And this is already the main issue: The larger n , the (exponentially) larger the number of possible solutions and thus, the larger the runtime of algorithms..

$\mathbb{N}_0$ the set of the natural numbers *including* 0, i.e., 0, 1, 2, 3, and so on. It holds that $\mathbb{N}_0 \subset \mathbb{Z}$.

$\mathbb{N}_1$ the set of the natural numbers *excluding* 0, i.e., 1, 2, 3, 4, and so on. It holds that $\mathbb{N}_1 \subset \mathbb{Z}$.

$\mathcal{NP}$ is the class of computational problems that can be solved in polynomial time by a non-deterministic machine and can be verified in polynomial time by a deterministic machine (such as a normal computer) ⁵².

$\mathcal{NP}$ -complete A decision problem is $\mathcal{NP}$ -complete if it is in $\mathcal{NP}$ and all problems in $\mathcal{NP}$ are reducible to it in polynomial time ^{52,101}. A problem is $\mathcal{NP}$ -complete if it is $\mathcal{NP}$ -hard and if it is in $\mathcal{NP}$.

$\mathcal{NP}$ -hard Algorithms that guarantee to find the correct solutions of $\mathcal{NP}$ -hard problems ^{23,26,72} have a worst-case runtime that grows exponentially with the problem scale. A problem is $\mathcal{NP}$ -hard if all problems in $\mathcal{NP}$ are reducible to it in polynomial time ⁵².

$\Omega(g(x))$ If $f(x) = \Omega(g(x))$, then there exist positive numbers $x_0 \in \mathbb{R}^+$ and $c \in \mathbb{R}^+$ such that $f(x) \geq c * g(x) \geq 0 \forall x \geq x_0$ ^{68,69}. In other words, $\Omega(g(x))$ describes a lower bound for function growth.

Glossary X



- $\mathcal{O}(g(x))$ If $f(x) = \mathcal{O}(g(x))$, then there exist positive numbers $x_0 \in \mathbb{R}^+$ and $c \in \mathbb{R}^+$ such that $0 \leq f(x) \leq c * g(x) \forall x \geq x_0$. In other words, $\mathcal{O}(g(x))$ describes an upper bound for function growth.
- $\Theta(g(x))$ If $f(x) = \Theta(g(x))$, then $f(x) = \mathcal{O}(g(x))$ and $f(x) = \Omega(g(x))$. In other words, $\Theta(g(x))$ describes an exact order of function growth.

$\text{quantile}_q^k(A)$ The q -quantiles are the cut points that divide a sorted data sample $A = (a_0, a_1, \dots, a_{n-1})$ where $a_{i-1} \leq a_i \forall i \in 1 \dots (n-1)$ into q equally-sized parts. $\text{quantile}_q^k(A)$ be the k^{th} q -quantile, with $k \in 1 \dots (q-1)$, i.e., there are $q-1$ of the q -quantiles. In the context of this book, define $h = (n-1) \frac{k}{q}$. $\text{quantile}_q^k(A)$ then can be computed as a_h if h is integer, i.e., $h \in \mathbb{Z}$, and as $a_{\lfloor h \rfloor} + (h - \lfloor h \rfloor) * (a_{\lfloor h \rfloor + 1} - a_{\lfloor h \rfloor})$ otherwise. It holds that $\text{quantile}_1^2(A) = \text{median}(A)$

$\mathbb{R}$ the set of the real numbers.

$\mathbb{R}^+$ the set of the positive real numbers, i.e., $\mathbb{R}^+ = \{x \in \mathbb{R} : x > 0\}$.

$\text{sd}(A)$ The statistical estimate $\text{sd}(A)$ of the *standard deviation* of a data sample $A = (a_0, a_1, \dots, a_{n-1})$ with n observations is the square root of the estimated variance $\text{var}(A)$, i.e., $\text{sd} A = \sqrt{\text{var}(A)}$.

$\text{var}(A)$ The *variance* of a distribution is the expectation of the squared deviation of the underlying random variable from its mean. The variance $\text{var}(A)$ of a data sample $A = (a_0, a_1, \dots, a_{n-1})$ with n observations can be estimated as $\text{var}(A) = \frac{1}{n-1} \sum_{i=0}^{n-1} (a_i - \text{mean}(A))^2$.

Glossary XI



- $\mathbb{X}$ The search space $\mathbb{X}$ is the space through which an optimization algorithm navigates in order to find good solutions. The search operators of metaheuristics are used to sample points $x \in \mathbb{X}$, for example. On many benchmark problems, the objective functions f directly evaluate the elements $x \in \mathbb{X}$, i.e., are defined as $f : \mathbb{X} \mapsto \mathbb{R}$. However, very often these elements x are not the same data structures that the user is interested in and/or that the objective functions evaluate: It is common that the search space $\mathbb{X}$ is a simple and plain data structure that can be tackled with standard operations or mathematical tools, such as real vectors $\mathbb{R}^n$, permutations, or bit strings, whereas the solution space $\mathbb{Y}$ can be something more complicated, like a Gantt chart. Then, the user only cares about the solutions $y \in \mathbb{Y}$ and not about the search space $\mathbb{X}$. A decoding $\gamma : \mathbb{X} \mapsto \mathbb{Y}$ is employed to translate the $x \in \mathbb{X}$ to $y = \gamma(x)$ with $y \in \mathbb{Y}$.
- x a point in the search space $\mathbb{X}$.
- $\mathbb{Y}$ the set of candidate solutions of an optimization problem is called the *solution space* $\mathbb{Y}$. The solution space corresponds to the data structure holding the candidate solutions y . These are the elements that are passed to the user after the optimization process is completed. They are also the parameters of the objective function(s) $f : \mathbb{Y} \mapsto \mathbb{R}$.
- y a candidate solution $y \in \mathbb{Y}$ is one possible solution to an optimization problem. After the optimization process is completed, one or multiple such solutions are passed to the user.
- $\mathbb{Z}$ the set of the integers numbers including positive and negative numbers and 0, i.e., $\dots, -3, -2, -1, 0, 1, 2, 3, \dots$, and so on. It holds that $\mathbb{Z} \subset \mathbb{R}$.